AN ALMA ARCHIVAL SEARCH FOR SPECTRAL TECHNOSIGNATURES TOWARD STARS WITHIN 40 PC

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ABSTRACT

We have searched 431 archival ALMA spectral windows, 89.6–495.1 GHz (Bands 3–8), toward 88 stars in 81 systems within 40 pc for unresolved spectral carriers, comparing each stellar position with 512 controls in a surrounding annulus. **No credible technosignature is found.** The sample is archive-target-complete but *epoch*-incomplete, only 102 of the 448 public blocks its targets hold being searched (§3). It holds 0.50 per cent of that volume's stars, over-representing F and G types and under-representing M dwarfs, so the null bears weakly on the habitable-zone question. Barnard's Star and Wolf 359 gain first millimetre technosignature limits. Four windows are stage-1 spatial outliers. Two toward β Pic and one toward HD 48370 match known circumstellar or foreground CO. The fourth, toward CP−72 2713, has no astrophysical attribution and is absent from a second, deeper block at the same tuning: no evidence for a persistent transmitter, though two blocks cannot exclude an intermittent one. The nominal 5σ trigger spans 2.3×10^{13}–6.8×10^{16} W EIRP over the 118 drift-resolved windows and 1.6×10^{13}–2.8×10^{17} W over the 313 unresolved-spectral-excess ones, which are 73 per cent of the sample and carry no drift discrimination. The effective unresolved-carrier threshold, the number to set against a transmitter power, is $\times 2.29$ higher under ALMA's spectral response, 5.4×10^{13}–1.6×10^{17} W and 3.7×10^{13}–6.4×10^{17} W; on the effective threshold no system here reaches Arecibo-class total power. Neither is a completeness limit: recovery is calibrated on one configuration and transfers only within a factor 0.5–2. The statistic and line mask were defined on these data. A held-out check on 271 later-searched windows finds the outlier rate as predicted but the stellar rank displaced towards the star, median 0.44 against 0.5, so the screen is mildly anti-conservative. The search is polarisation-blind.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

A technological civilisation, signalling deliberately or going about its affairs, may imprint the electromagnetic spectrum detectably, a *technosignature* (Sheikh 2020; Vidal et al. 2026). Six decades of radio searches cover many thousands of stars yet sample only 6×10^{-18} of the haystack of transmitter frequency, bandwidth, power and duty cycle (Wright et al. 2018). Three gaps persist. The millimetre and submillimetre regime is very sparsely searched, and everything above that axis's ∼115 GHz upper limit more sparsely still. Most surveys are aimed at interest-selected classes of star, leaving the local stellar population unsampled as a population. Few reprocess *existing* interferometric archives.

This paper addresses all three, asking how strongly existing ALMA observations constrain unresolved artificial spectral emission from nearby stellar systems. The sample it uses characterises the stars ALMA has observed, over-representing F and G stars and under-representing M dwarfs (§3), and every limit inherits that bias. We mine the archive, correcting for proper motion and Doppler drift, for unresolved spectral carriers (channel-confined excess at native ALMA resolution) and anomalous millimetre continuum. Throughout, "narrowband" describes the assumed intrinsically narrow transmitter and never the resolving power of the data (§4). Channelisation sets which of two search classes a window belongs to, and the completeness of each class is measured in §4.5. Three contributions follow. The first is the systematic exploitation of archival ALMA stellar fields, 107 target/band datasets mined from public data with no new observing time. The second is spatial-control null calibration for interferometric SETI, in which the same statistic is evaluated at the star and at 512 control positions so that the false-alarm rate is measured and not assumed. The third is a quantitative accounting of the frequency, power and drift domain actually searched, which gives the non-detection explicit boundaries. No ancillary analysis carries a technosignature disposition. One worked case runs through the paper, a crossing towards β Pictoris, which is the largest star-versus-control contrast here and is attributed to the disc's known circumstellar CO across two transitions and three epochs (§5.3). Figure 2 is the roadmap: it gives the selection from catalogue to searched sample on the left, and the decision chain from window to disposition on the right.

The statistic and the molecular-line mask were both defined on these data, and the statistic once revised (§5.4), so the rank probabilities and false-alarm figures describe this dataset, do not independently validate the pipeline's

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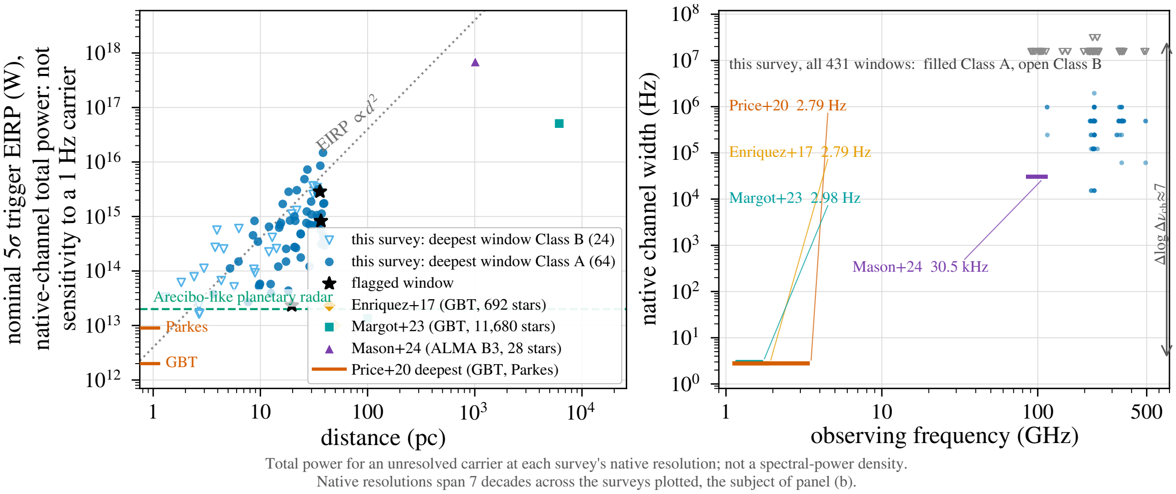

FIG. 1.— Literature context of the 5σ trigger thresholds, in total power (§2). *(a)* EIRP against distance. Blue: this survey, deepest window per star (88 stars, 1.30–39.63 pc; filled: Class A, drift-resolved; open: Class B, unresolved excess); orange: Enriquez et al. (2017); aqua: Margot et al. (2023); violet: Mason et al. (2024); left-edge ticks: deepest per-target limits of Price et al. (2020). Grey dotted: EIRP $\propto d^2$ at constant received flux. Green dashed: the Arecibo 2.38-GHz EIRP benchmark, a power scale only (§4.3). *(b)* Native channel width against observing frequency for the same windows and programmes: native resolutions differ by $\sim 10^7$, so the panels should be read together and a lower EIRP here does not imply greater sensitivity to a hertz-wide carrier; conventions differ between surveys.

false-positive rate, and are exploratory rather than pre-registered. An out-of-sample check on windows searched after both were frozen is in §5.3.

2. BACKGROUND

Radio searches, from Cocconi & Morrison (1959) and Drake (1960) through the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002), divide into wide-field commensal and targeted surveys (Zhang et al. 2020; Czech et al. 2021; Morrison et al. 2023; Li et al. 2026; Johnson et al. 2023; Valente et al. 2025; Tremblay et al. 2024), with machine-learning interference rejection now central (Ma et al. 2023; Choza et al. 2024; Jacobson-Bell et al. 2025; Poznanski et al. 2025; Garrett et al. 2026), a capability our classical threshold search lacks. Essentially all of that work lies below $\sim$10 GHz, and the millimetre regime above it has been searched once, with ALMA (Mason et al. 2024). There are physical reasons to go higher, since interstellar scattering and dispersion fall steeply with frequency, a given aperture delivers a tighter beam per watt of EIRP, and directional communication, radar and beamed power are already engineered as coherent mm-wave emission. Atmospheric transmission, receiver noise, photons per watt per hertz, and pointing and phase stability raise the cost of such a search without forbidding it.

Mason et al. (2024) searched archival Band 3 windows toward 28 "stellar bycatch" stars, observed for other science and usable for free, placing $\mathrm{EIRP_{min}} > 7 \times 10^{17}$ W for the nearest and identifying the challenges of high-frequency work: drift rates scale with frequency, smearing erodes sensitivity without a drift search, and the dense forest of molecular lines raises the confusion background. Our pilot (White 2026) extended that methodology to TRAPPIST-1 across Bands 3, 6 and 7 ($\mathrm{EIRP_{min}} \sim 5$–$10 \times 10^{13}$ W, no detection) and built the ranked 100-star list this survey grew from. Nothing here depends on it, because every claim attributed to it below is restated from the products released with this work. The three contributions listed in §1 are new, to our knowledge, *in an ALMA archival technosignature survey.*

Every window here was re-reduced and re-searched from the raw archive data with the pipeline of §4, so where a star appears in both, TRAPPIST-1 included, the numbers to cite are this paper's frozen products.

Figure 1 places the nominal EIRP thresholds among published searches in *total power*. As the in-panel note warns, vertical position carries no technosignature merit. This survey's competitive EIRP toward the nearest systems comes predominantly from the d^2 factor (d spans a factor $\sim 10^3$ in d^2), and only weakly from sensitivity to Hz-wide carriers, to which a 15.625-MHz channel and a 2.8-Hz SETI channel respond very differently (§4).

A non-detection must become a quantitative exclusion statement: Wright et al. (2018) formalised the "Cosmic Haystack" search volume, Margot et al. (2023) its conversion into an upper bound on the transmitter-hosting fraction, and Sheikh et al. (2025) asked where Earth's technosignatures would be detectable. That bound is valid only if the sample represents the underlying population, which interest-selected samples do not. Most surveys select by proximity, spectral type, planets or Earth Transit Zone membership; the ALMA-specific form, disc selection, skews the sample young and dynamically active. We instead take every star within 40 pc lying in the usable field of at least one public ALMA observation (§3), a sample archive-target-complete for the observed subset of the local population in a fixed volume while making no completeness claim for that population itself. That choice removes technosignature-specific selection and leaves the archival astrophysical selection in place; the 168 sample entries are $\sim$1 per cent of the catalogued population within 40 pc (the 88 actually searched here are $\sim$0.5 per cent), and the residual coverage selection is characterised as a completeness function (§3, Table 4).

3. SAMPLE SELECTION

TABLE 1
Nomenclature for search output and the two senses in which a probability is quoted.

Term	Meaning
crossing	one channel×drift cell at the stellar position with $T \geq 5$: a threshold crossing (state 1); counted per window in the main text and per cell in Appendix G
stage-1 spatial outlier	a window whose $T_\star$ exceeds all 512 control statistics ("star-exceeds-ring"), a screening device that carries no candidate significance (state 2); shortened to "stage-1 outlier" after first use
$p_{\mathrm{rank,min}}$	$1/(N_{\mathrm{ctrl}}+1) = 1/513$, the rank *resolution* of one window's control ensemble: the smallest per-window probability the design can express
P(false flag)	the survey-level false-positive probability after thresholding, correlation and non-Gaussianity. The two differ, the first being a property of one window's ensemble and the second of the whole survey, and the survey-wide Bonferroni scale 1.2×10^{-4} lies a factor 16 below the first (§5.3)
archive-target-complete	every star the archive-selection procedure of §3 identifies as covered is searched
epoch-incomplete	only 102 of the 448 public execution blocks those targets hold are searched

The sample is archive-*target*-complete but epoch-incomplete within 40 pc, a much weaker condition than volume completeness. Table 1 defines both terms. The procedure admits every star within 40 pc of the Sun (Gaia DR3 parallaxes, distances direct inversions $d = 1/\pi$; every entry $\pi \geq 25$ mas with a parallax measured to better than 20 per cent, which is the condition the archive query applied) whose proper-motion-corrected position at the observation epoch falls in the quoted field of view of at least one public ALMA execution block. The star need not sit at the pointing centre, and §5.2 shows the quoted field of view admitted 53 stars no observation in fact contains, so "covered" is a property of our selection and not a guarantee. Disc, planet and spectral type enter as descriptive metadata only. Units are defined in §4.1. This gives 168 unique target stars, processed nearest-first. Because trigger thresholds scale with d^2 at fixed noise, that order is sensitivity-ranked, so the prevalence statements of §6 are conditional on it.

Composition follows from the archival selection and is tabulated against the reference census in Table 8. The sample is F- and G-enriched and deficient in M dwarfs, and 39 of the 431 windows come from 3 systems; §6.3 gives the figures and their cost. The spectral-class proxy is `teff_gspphot`, available for 52 of 88 sample stars and missing preferentially for the coolest objects, so the M fraction is a lower limit and the earlier-type ratios upper limits. The debris-disc hosts contribute dusty continua, kept from masquerading as anomalies by the continuum lane's screen (§4).

The 168 stars are complete against, and conditional only on, the 40-parsec reference catalogue intersected with the public-archive snapshot of 2026 September 8. Against that Gaia DR3 catalogue of 17 566 stars they are $\lesssim$1 per cent and the 88 searched are 0.50 per cent, fewer still once §5.2 removes those never observed (Fig. 2). The two parallax cuts are not interchangeable, the sample's binding condition being $\sigma_\varpi/\varpi < 20$ per cent against the census's 10 per cent, and the realised sample is better than its condition, reaching 2.48 per cent at worst. Because the census is incomplete at both ends, the fraction shows only that coverage is small.

Stellar counts throughout are catalogue-entry counts. The 88 searched entries comprise 81 independent systems, differing by seven designation-linked component pairs (SIMBAD designations; per-entry rows, distances and unresolved-pair notes in the machine-readable catalogue). The 107 star/band datasets, 431 windows and 102 execution blocks therefore derive from 81 systems. The 431 rows comprise 396 distinct (execution block, spectral window) datasets, and the larger number is used as the trials denominator throughout, which is conservative.

"Usable public ALMA coverage" is a five-criterion decision tree, frozen absent an erratum: (i) at least one public execution block (EB) whose visibilities have passed ALMA's second-stage quality assurance (QA2) and cover the star's epoch-propagated position, with no minimum integration, sensitivity, antenna count, mosaic-tile count or array configuration beyond QA2; (ii) that position inside the field's primary beam, the extraction aborting beyond $2\,\theta_{\mathrm{PB}}$ and the largest retained offset being $0.46\,\theta_{\mathrm{PB}}$, so nothing approaches the bound; (iii) each spectral window admitted individually, on QA2 vetting alone; (iv) no programme-category restriction, science, calibration and observatory projects entering on identical terms; and (v) a fixed archive snapshot date (2026 September 8). No star or window was excluded by an unlisted criterion, and none for short integration.

Target-completeness holds in targets and tunings. It does not extend to epochs or to integration time, and the shortfall is quantified here. The 102 member observing unit sets (MOUS) that supply the searched blocks hold 448 public execution blocks between them, of which 102 were searched (22.8 per cent) and 346 were not. Two deliberate choices cause that, a per-target download budget capping the blocks taken at 3, and spectral-window de-duplication by identifier across the blocks of one set. 75 of the 102 sets hold at least one unsearched block, supplying 304 of the 431 windows (71 per cent) and 62 of the 88 stars, for which up to 34 public blocks exist against the 3 this survey searched. Every epoch statement here, the epoch-activity bound of Appendix I included, therefore concerns searched epochs, and the archive would support a stronger one. 67 of those blocks have since been searched, as the recurrence tests of §5.3 and as the out-of-sample check of §5.3. They sit outside this frozen release, so the counts in this section and the trials budget of §5.3 are unchanged by them.

ALMA project metadata quantify pointed versus serendipitous coverage: almost every spectral-window-covered star is the pointed science target in at least one execution block, leaving Wolf 219 the sole serendipitous star, so the residual selection function of an archive-target-complete sample is the union of ALMA target lists.

4. DATA AND METHODOLOGY

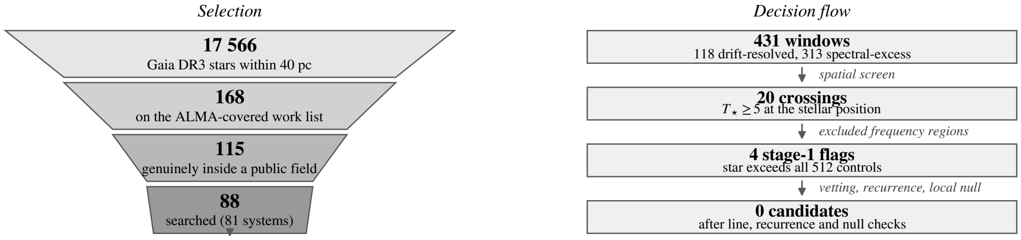

Fig. 2.— Left: the selection funnel. The three denominators the text keeps apart are the catalogued population, the 168-entry ALMA-covered work list, the ∼115 stars a public field genuinely contains, and the 88 searched. Right: the decision flow from spectral window to disposition. The 512-control comparison is a candidate-generation screen, and the gates in italic are where a crossing can leave the chain.

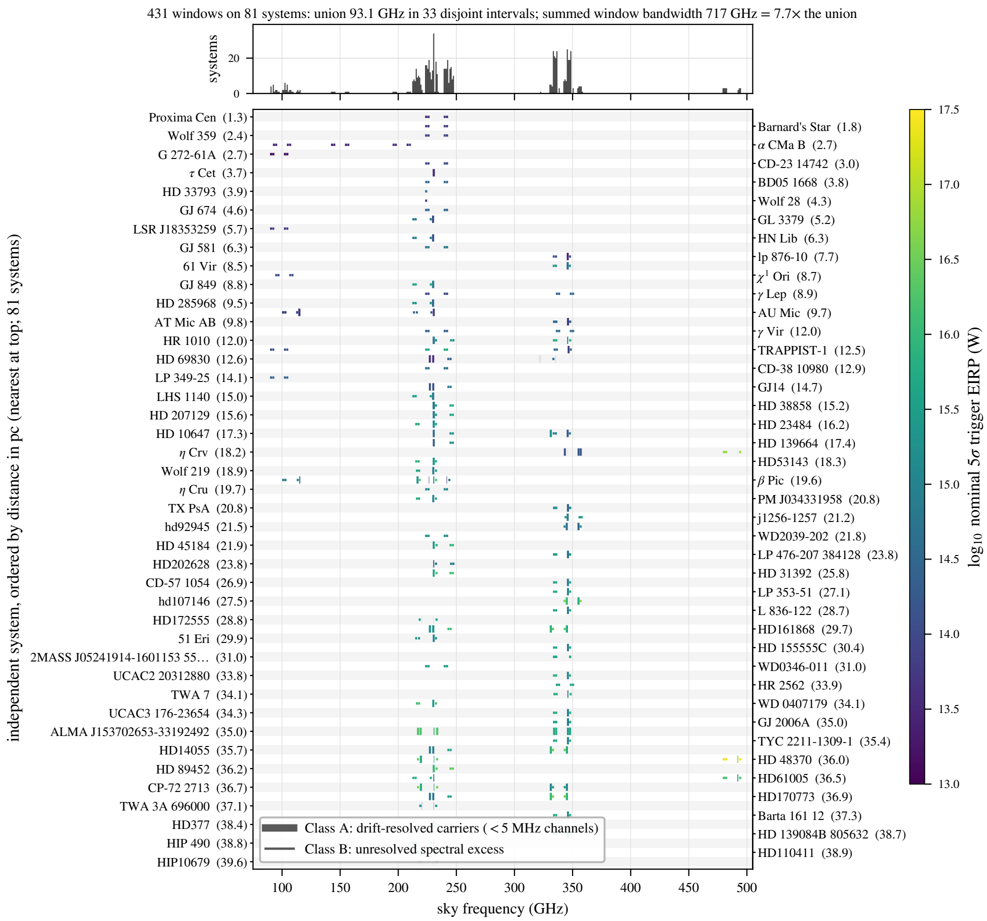

Fig. 3.— What this survey sampled in frequency. Each searched window is a horizontal segment on its host system's row, rows ordered by distance and labelled alternately on the left and right axes, coloured by nominal $\mathrm{EIRP}_{5\sigma}$, with line weight distinguishing the drift-resolved (Class A) from the unresolved-spectral-excess (Class B) channelisation; the upper panel counts systems per gigahertz of sky frequency. The coverage is narrow, clustered and repeatedly revisited: 33 islands concentrated near 230 and 345 GHz, with summed window bandwidth 7.7× the union, which is what the in-band transmitter fractions of Appendix I condition on (Scope of the constraints, §4).

The design is one shared archive-ingestion stage feeding two independent search lanes, and Table 2 gives the data path of the primary lane in order. For each target we mine, within a bounded budget, the calibrated archival execution blocks covering the star's Gaia-propagated position. Their visibilities feed (i) the *primary* lane, a Doppler-drift-corrected spectral-carrier search following Mason et al. (2024) and Enriquez et al. (2017), which converts each non-detection into an equivalent isotropic radiated power limit $\mathrm{EIRP}_{5\sigma} = 4\pi d^2 S_{\min}\Delta\nu$. Here $S_{\min}$ is five times the searched window's per-channel rms, primary-beam corrected, $\Delta\nu$ the native channel width and d the target distance (worked numerically in Appendix A). Five terms enter their systematic budget. They are the ALMA absolute flux scale, 5–10 per cent and band- and epoch-dependent; residual atmospheric phase after water-vapour correction and phase referencing; the primary-beam model at the stellar offset; the instrumental spectral response, median ×2.29, which is the largest of the five and is carried explicitly as P_{eff} by the boxed rule below; and visibility calibration. Each multiplies $S_{\min}$, and hence $\mathrm{EIRP}_{5\sigma}$, directly; *Gaia* distance errors are negligible. The atmospheric term is a coherent point-source loss $\exp(-\sigma_\phi^2/2)$, typically 5–20 per cent at 230–345 GHz. It is common to the star and the annulus within the isoplanatic patch, so it is no threat to exchangeability. The same visibilities feed (ii) an *ancillary* line-excluded continuum lane

(Appendix F). In most windows an Hz-wide carrier is unresolved within a 15.625 MHz channel, so the data cannot show a one-channel excess to be intrinsically narrow. The search is reported as two survey classes, distinguished by the dimensionless drift parameter

$$\eta_{\rm drift} \;=\; |\dot{\nu}_{\rm max}|\,T\,/\,\Delta\nu_{\rm ch}, \tag{1}$$

the channel-width fraction the *maximum* searched drift $|\dot{\nu}_{\rm max}|$ traverses over the full on-source track T at native channel width $\Delta\nu_{\rm ch}$. The track is the right time-scale, the de-drift step summing along it; $\eta_{\rm smear}$ (Appendix A) is the same ratio over one integration. A window is *drift-resolving* when $\eta_{\rm drift} \gtrsim 1$, carrying a carrier across enough channels for its motion to be measured. The drift ceiling inside $\eta_{\rm drift}$ is a search-grid boundary, the widest trial drift the stack evaluates, not a prior on transmitter drift rates; carriers drifting faster are missed, and §6.1 gives the resulting selection function (Fig. 8).

Class A is the drift-resolved spectral-carrier search: the 118 windows of fine channelisation, native widths 15.259–1953.125 kHz (median 488.3 kHz; 62 per cent at the 488.3-kHz mode), spanning $\eta_{\rm drift} = 0.34$–422 (median 12.8). It is the only class with measured end-to-end recovery of a drifting carrier. *Class B* is the unresolved spectral-excess search: the 313 coarse windows, 73 per cent of the sample, native widths 15.625–31.25 MHz (309 windows at the 15.625-MHz mode), spanning $\eta_{\rm drift} = 0.004$–1.35 (median 0.36). It is sensitive to persistent near-stationary excess power, carries no drift discrimination, and is never described as a narrowband search anywhere in this paper.

$\eta_{\rm drift}$ is the physical criterion and the classes are labelled by the instrument. Channelisation is bimodal, with no window between 1953.125 kHz and 15.625 MHz, an empty gap of factor 8.0, so any boundary inside the gap assigns identical membership and figures label the classes by name. The two criteria do not coincide exactly, and the overlap is quantified: 12 long-integration Class B windows reach $\eta_{\rm drift} \geq 1$, and only marginally (maximum 1.35), where a one-to-two channel traverse supports no discriminative drift fit after baseline removal; 6 short Class A windows fall below unity and are conservatively kept in Class A. Each window's $\eta_{\rm drift}$, $\eta_{\rm smear}$, trial drift count and maximum searched acceleration $a_{\rm max} = 3.6$–$4.0\,{\rm m\,s^{-2}}$ are columns of the released catalogue, so any reader may reclassify on the physical criterion alone. Results are given per class first and combined only where statistically appropriate (§5.3).

Both lanes exclude catalogued molecular transitions at tolerances suited to their statistics. The continuum lane masks the common species at native resolution before integrating; the carrier lane applies a $\pm 50\,{\rm km\,s^{-1}}$ stellar-frame velocity exclusion mask (§5.3) to any threshold crossing. That half-width was widened after the β Pictoris crossings had been seen, so the safeguard belongs here rather than in a later section: reclassifying all 20 crossings at ± 20, ± 30, ± 50 and $\pm 100\,{\rm km\,s^{-1}}$ admits no new unattributed outlier, and at every width CP-72 2713 remains the only unmasked one. The mask is *excluded search space*: the unmasked survey covers 93.1 GHz of unique sky frequency, the effective default quoted throughout is the $\approx$88.2 GHz surviving it, and no constraint is placed on carriers inside the excluded velocity regions. Masking reduces the frequency space a candidate can occupy while leaving the per-channel noise untouched, so no threshold moves. All flux densities and EIRP thresholds are total intensity (Stokes I), summed over the two parallel-hand cross-correlations with QA2 weights; no cross-hand extraction or circularity test was performed, so the construction is polarisation-blind in amplitude and a strongly polarised artificial signal responds like an unpolarised one at fixed total power. The estimator is safe in this respect, though the data have not been audited for it. Both parallel hands are delivered for 102 of the 102 searched execution blocks, which is all of them, so the discrimination described next is available from the existing data and needs no new observation. What the retained products do not record is per-correlation flagging inside QA2, so we cannot say what fraction kept both hands unflagged all the way through; one that lost a hand would be $\sqrt{2}$ less sensitive than its threshold implies, with nothing here to catch it. What Stokes I costs is discrimination, not sensitivity: an artificial coherent carrier is more likely to be strongly polarised than the unpolarised line and dust emission that fills this archive, so a polarisation measurement would separate the two for free.

How to read every limit in this paper. *Three quantities are kept apart throughout, and only the first is what the pipeline acts on.*

The ***trigger power*** $P_{\rm trig}$ *is the nominal EIRP*$_{5\sigma}$*: five times the searched window's per-channel rms, at that window's native channel width, for a transmitter at the star's position matching the searched morphology. It is the level at which the search fires, it is what every "*5σ*" below means, and it is what the figures label as the nominal trigger.*

The ***effective unresolved-carrier threshold*** $P_{\rm eff}$ *is* $P_{\rm trig}$ *corrected for the instrument's spectral response. ALMA applies online Hanning smoothing, which leaves only 0.38 to 0.50 of a sub-channel tone's power in the peak channel depending on where in the channel it falls, median 0.44 (Appendix A). Nominal thresholds are therefore optimistic by* $\times 2.00$ *to* $\times 2.67$*, median* $\times 2.29$*, and* $P_{\rm eff} = 2.29\,P_{\rm trig}$ *is the number to compare with any transmitter power. It is analytic, from the* 0.25/0.5/0.25 *kernel, and window-specific:* $\times 2.29$ *where the archive reports an effective resolution of 2.00 channel spacings (403 windows) and* $\times 1.14$–1.33 *for the 28 with online channel averaging. No injection measured it, the injected tones being unsmoothed (Appendix A): it is a sensitivity correction and not a measured completeness. It is not a small correction and we treat it as the principal physical threshold.*

The ***completeness limit*** P_x*, the power at which measured recovery reaches* x *per cent, is a different quantity again and is determined here for one configuration only:* $P_{50} = 1.10\,P_{\rm trig}$ *and* $P_{90} > 2\,P_{\rm trig}$ *on the calibration window's drifting class (Appendix B). Transferring that curve to a window observed in another band, channelisation or integration time is an assumption, bracketed at* 0.5–2$\times$ *(Appendix I); elsewhere* P_{90} *and* P_{95} *are* not determined. *What this survey has is a measured reference recovery function of incompletely established transferability, and not a measured Class A* *completeness function.*

Every headline EIRP is quoted beside its native channel width, and the machine-readable catalogue carries $P_{\rm trig}$ *and both response-corrected values per window.*

Three signal morphologies are detectable here, and §6.1 quantifies each against real windows: a continuous frequency-stationary carrier, a continuously drifting one, and an intermittent one.

4.1. *Statistical units and independence*

Six units appear below and they are not interchangeable, so we fix them here and say which one carries which probability. A *catalogue entry* is one star as SIMBAD designates it, 88 of them searched; a *physical system* is the set of entries gravitationally bound together, 81 of them, seven designation-linked pairs accounting for the difference. A *target/band* dataset is one entry in one ALMA band, 107 in all. An *execution block* is one archival observation, 102 searched, and it is the unit that shares weather, calibration and correlator setup. A *spectral window* is one window of one execution block: 431 catalogue rows, which reduce to 396 distinct (execution block, window) datasets once repeated archive products are collapsed. An *independent frequency trial* is neither of those: within a window the channel×drift grid is oversampled on both axes, so its 177–6 866 208 cells carry about a factor 5.3 fewer independent trials than cells (Appendix G). An *independent epoch* is a separate execution block at the same tuning, and blocks of one scheduling block executed on one night are not independent epochs in any useful sense.

Which unit enters a population-level statement therefore depends on the statement. Per-window rank probabilities are conditional on the window and use no denominator at all. The survey-level flag expectation uses 431 windows, which is conservative because the 396 distinct datasets are the smaller number and because windows within a block are correlated; **431 windows are not 431 independent trials**, and no probability here should be read as though they were. Anything about how common transmitters are must use 81 systems, never 88 entries and never 431 windows. Anything about persistence uses epochs, of which most stars here have one.

4.2. *The search statistic, and the words used for its output*

The primary statistic is formed on extracted spectra, never on images or visibilities. It operates on the QA2-calibrated measurement set as delivered, per integration and per native channel. The archive's flagging carries through untouched and no spectral regridding or interpolation is applied (Table 2, step 3). What the extraction returns at a propagated sky position $\mathbf{x}$ is the real part of the weighted vector average of the phase-shifted visibilities, which is the natural-weighted dirty-map value there. There is no deconvolution, so a quoted $S_{\rm peak}$ in a field with extended emission is a dirty-map amplitude. Baseline removal follows extraction (step 4). The primary-beam correction enters the quoted thresholds and fluxes (§4.3) and leaves untouched the extracted amplitudes the statistic ranks. For a sky position $\mathbf{x}$, each per-integration spectrum $I(\mathbf{x}, t, \nu)$ is baseline-subtracted by a block median along the *frequency* axis. No temporal median is applied at any stage (§4.4). The residual dynamic spectrum is stacked along each trial linear drift rate $\dot\nu$ with inverse-variance weights, and the statistic is the maximum of that stack over the channel×drift grid,

$$T(\mathbf{x}) = \max_{\dot\nu,\,\nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot\nu t)\,/\sigma^2(t,\nu)}{\left[\sum_t 1/\sigma^2(t,\nu)\right]^{1/2}}, \tag{2}$$

a matched filter whose denominator is the noise of the stacked spectrum. Only when σ is the same for every integration and channel does it reduce to a single-channel amplitude divided by one noise scale; otherwise the weights vary along the track. The scale $\sigma(t,\nu)$ is 1.4826× the median absolute deviation taken *across the control positions* at each integration t and each channel ν, which makes it global in position, local in frequency and in time, and shared by the star and by every control.

The across-position median enters only in forming that deviation and is never subtracted from the numerator, so T is an absolute stacked signal-to-noise carrying a cross-position noise scale, and it measures no local contrast. A line filling the primary beam therefore drives star and annulus up together instead of cancelling.

The baseline filter takes the median in contiguous blocks of 65 channels and interpolates linearly between block centres, using $\max(2, n/65)$ blocks over n usable channels. Its effective width is therefore 65 channels in the 118 windows long enough to hold more than two blocks, and $n/2$ in the short coarse windows where the block count collapses to two. This sets a ceiling on the linewidth the filter passes: ≈32 MHz at the fine class's modal channelisation, and ≈0.9 GHz at the coarse class's (Appendix A). The crossing channel is *not* excluded from the scale. The median is taken across positions at a fixed channel, so what matters is how many of the 512 controls carry the feature. A compact feature at the star appears in none of them and the 50-per-cent breakdown point holds; emission filling the primary beam appears in all of them, which is the HD 48370 case (§5.3). $T_\star \equiv T(\mathbf{x}_\star)$ at the proper-motion-propagated stellar position, and the same operation, drift grid, channelisation and noise map give $\{T_i\}$ at all 512 control positions.

The controls are placed by `probe_positions()` in the released extraction code, which adds its offsets to the star's own direction cosines, so the annulus is centred on the *star* itself, wherever the star sits relative to the field phase centre. It holds 512 positions drawn uniform in area, $r = \sqrt{U(r_{\rm in}^2, r_{\rm out}^2)}$, and uniform in azimuth, from the fixed seed 20260825 re-applied per window. All windows therefore share one pattern in units of $\theta_{\rm PB}$, and two windows of one

TABLE 2
THE STATISTICAL DATA PATH, IN ORDER. STEPS 1–8 RUN IDENTICALLY AT THE STELLAR POSITION AND ALL 512 CONTROLS; ONLY STEP 9 ONWARD DISTINGUISHES THEM.

1	calibrated measurement set from the ALMA archive
2	stellar position propagated to the observation epoch (Gaia DR3)
3	spectral extraction at that position, native channelisation
4	per-integration baseline removal: block median along frequency ($W = 65$)
5	stack along each trial linear drift rate
6	robust noise map $\sigma(t, \nu)$: MAD across the 512 controls
7	$T(\mathbf{x})$ from Eq. 2, an inverse-variance-weighted stack
8	repeat 3–7 at 512 control positions on a ring
9	threshold: $T_\star \geq 5$ (a crossing)
10	rank test: $T_\star > \max_i T_i$ (stage-1 spatial outlier)
11	molecular-line mask, $\pm 50\,\mathrm{km\,s^{-1}}$ stellar frame
12	vetting: recurrence, closure phase, cross-target occupancy

tuning sample identical sky positions (Fig. 5). Radii run 0.14–0.78 $\theta_{\rm PB}$, with $\theta_{\rm PB}$ evaluated at each window's own median frequency, so the annulus scales window by window. A separate inner check-ring of 135 positions serves the point-source test, and the full per-window geometry ships as catalogue columns.

The frozen products carry none of that geometry, so it is reconstructed from the code, and two checks confirm the layout that ran: the inner check-ring it generates has the 135 positions the retired region-max statistic maximised over, and where extended CO fills the beam the control statistics fall with reconstructed radius, at Spearman $\rho = -0.49$ in the HD 48370 Band 6 window. Neither would survive a wrong position-to-index mapping.

Placing controls off-centre costs sensitivity. A star near the pointing centre sits at primary-beam response $\approx$1 while the annulus samples 0.95 at 0.14 $\theta_{\rm PB}$ falling to 0.19 at 0.78 $\theta_{\rm PB}$, area-weighted mean 0.47. Star and controls are therefore not exchangeable in flux density, and they remain exchangeable in the signal-to-noise ratio the statistic actually ranks, because $T(\mathbf{x})$ is formed on uncorrected amplitudes against one shared noise scale. Extended sky brightness does not cancel: smooth emission lives on short baselines whose phases barely change under a rotation of a few arcsec, so star and annulus rise *together*, as they do for HD 48370, while emission offset from the star raises particular controls above it, as in the same star's ^{13}CO window (ring maximum 20.4 at 9.2 arcsec against 4.6 at the star). Two asymmetries remain and they run in opposite directions: a suppressed control can only make a star-exceeds-ring outcome more likely under the null, whereas a multiplicative calibration error scales with the continuum flux at the extracted position and so favours the star, where that flux is not zero.

Neighbouring controls are correlated below the synthesised-beam scale, so a window's annulus carries $N_{\rm eff} \simeq 30$–43 independent spatial trials in the compact configurations where the symmetric reprocessing runs measure it. That is the worst case, the annulus containing far more resolution elements than the 512 positions drawn in all but those configurations (Appendix G). The rank test needs exchangeability rather than independence in any case, and correlation among controls costs resolution without touching the 1/513 floor. The frozen products do not carry the synthesised beam or the array configuration, so radii are quoted in arcsec and in $\theta_{\rm PB}$; the ALMA archive carries both, and what they say about this geometry, including the 33 per cent of windows taken with the ACA 7 m array against a $\theta_{\rm PB}$ defined on 12 m, is in Appendix G.

Ranking $T_\star$ against 512 values fixes the smallest per-window p-value the design can express at $1/513 = 1.95 \times 10^{-3}$, while the survey-wide Bonferroni scale over 431 windows is $1.2 \times 10^{-4} = 0.05/431$, smaller by a factor of 16 (Appendix G). **No single event in this design can therefore reach survey-wide significance.** The 512-control comparison is a candidate-generation screen and never a significance test: a stage-1 spatial outlier selects a window for examination and establishes nothing about it. Table 1 fixes the vocabulary and Table 2 allows the statistic to be reimplemented from the main text alone.

Residual continuum after baseline removal, self-calibration residuals and position-correlated calibration error all concentrate at or near the phase centre, where the pointed science target sits, and σ is built from the annulus and carries none of it. These stars are not all blank there, the continuum lane detecting 17 of 57 target/bands (Appendix F). A multiplicative spectral calibration error carries the same asymmetry in the unfavourable direction, since a residual bandpass ripple of fractional amplitude ϵ leaves a channel-confined artefact of order $\epsilon\, S_{\rm cont}$ at the extracted position and essentially nothing on the annulus. The direct test of the premise is §5.3, which compares the star's own residual variance with the controls' in the four flagged windows and finds them equal to better than a per cent, and two indirect ones agree with it. Recomputing the scale locally, within a band around each feature and excluding the feature itself, gives local-to-global ratios of 0.97 (CP−72 2713), 1.29 (HD 48370), 0.93 (β Pic B3) and 1.02 (β Pic B6): the global scale is accurate to a few per cent except towards HD 48370, where the field is filled with CO and rescaling both the statistic and its ensemble leaves the window flagged ($T_\star = 21.0$ against 20.8), which is the disposition it already has. And because the same estimator runs at the star and at every control, a badly misstated σ would displace the control distribution, which the uniform pseudo-star ranks exclude. None of this reaches a per-crossing local scale inside the search, which remains an open item (§6.4).

A rank means something only when the identical operation runs at the star and at every control: under a pure-noise null a region maximum over $n_{\rm src}$ positions at the star exceeds all $n_{\rm ctrl}$ single controls with probability $n_{\rm src}/(n_{\rm src}+n_{\rm ctrl}) =$ 20.9 per cent for the 135-position check-ring, whatever the window and whatever the noise distribution. The released statistic therefore evaluates one position at the star and one at each control; §5.4 dates the earlier asymmetric version and audits the windows it affected.

The searched drift-rate range is the larger of (a) a generic frequency-scaled default (12–13 Hz s^{-1} GHz^{-1}), the Mason et al. (2024) fiducial with a threefold safety margin, and (b) for confirmed exoplanet hosts, the maximum line-of-sight orbital acceleration of the innermost known planet, so a transmitter on a short-period planet cannot drift out of the window. Drift rate maps onto line-of-sight acceleration through the first-order Doppler relation

$$a_{\rm los} \;=\; c\,\frac{\dot{\nu}}{\nu}, \tag{3}$$

so the ceiling is quoted in cross-band-comparable form $\dot{\nu}/\nu$: 12–13 Hz s^{-1} GHz^{-1} is $(1.2\text{–}1.3) \times 10^{-8}\,{\rm s}^{-1}$, that is $|a_{\rm los}| = 3.6\text{–}3.9\,{\rm m\,s^{-2}}$ independent of observing frequency, the acceleration at $\sim$0.04 au around a solar-type star. Earth-motion contributions lie two orders of magnitude below and are subsumed by the grid. The grid is uniform in $\dot{\nu}$ over $n_{\rm drift}$ steps (2–1944 across windows, recorded per window), so a carrier midway between trial rates leaves a residual drift traversing at most 0.26 of a native channel over the longest fine-class track, and the resulting amplitude loss is an order below the recovery slope measured between 5σ and 10σ. Two limits of the model matter more. The rates searched are *apparent* topocentric ones, so the survey does not separate a deliberately chirped carrier from an accelerating platform and constrains only linear apparent drift inside the ceiling; and an emitter need not be tied to a known planet at all, a free-flying platform or rotating structure accelerating as it chooses and falling outside these constraints without being excluded by them. For circular orbits the ceiling covers a transmitter co-orbiting an Earth analogue by three orders of magnitude and Proxima b comfortably, while TRAPPIST-1 b ($a_{\rm los} = 4.00\,{\rm m\,s^{-2}}$) sits at it and c ($2.14\,{\rm m\,s^{-2}}$) reaches it at modest eccentricity, which raises the periastron value by up to $(1{+}e)/(1{-}e)^2$ (Fig. 8).

Across the 431 windows maximum searched drift rates span 1.1–5.9 kHz s^{-1} and trial drift rates per window 2–1944 (median 4). The threefold margin matters. At 1× the search would have been blind to two of its threshold-crossing windows, including the AU Mic crossing, while at 5× no disposition changes; transmitters drifting faster than the 12.0–13.3 Hz s^{-1} GHz^{-1} ceiling remain unconstrained. Intra-integration smearing is negligible for 422 windows: at most 1.10 channels, median 0.001. 9 windows have $\eta_{\rm smear} < 0.99$ (Appendix A); 5 of them, all 15.3 kHz Band 6 windows and so among the most drift-capable here, carry effective thresholds 1.65–1.74× nominal, and 2 are stage-1 outliers. Channel widths are effective resolutions from the measurement sets' spectral window tables (Appendix A).

4.3. *Frequency and velocity reference frames*

Every frequency axis here, searched spectra, tabulated ranges and crossing frequencies alike, is *topocentric* as delivered by the ALMA correlator, with no velocity-frame shift applied at any stage. Velocities enter only in the vetting stage's line mask, whose conversion chain is in Appendix A. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, and not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold, and the commonest confusion when limits are compared across surveys. Channel dilution spreads a sub-channel carrier's total line flux across the channel, so the minimum detectable total power is $4\pi d^2 S_{\rm min}\Delta\nu_{\rm ch}$, independent of the transmitter's own width below the channel (Appendix A). EIRP$_{5\sigma}$ is therefore the minimum *total* power of a channel-confined carrier and never a spectral power density. The 1-Hz and 3-Hz limits of Hz-resolution surveys are total powers in the same sense, each the best case of its own channelisation, which is what makes the axes of Fig. 1 commensurable. Channelisation is the dominant price the archive exacts. We do not project an Hz-resolution sensitivity from it, because scaling by $P_{\rm min} \propto \sqrt{\Delta\nu}$ across six orders of magnitude in spectral resolution would be a thought experiment and not an achievable ALMA sensitivity.

Every spectral window of an observation enters the spectral-carrier search, capped at 8 per target to bound compute cost; the cap bound one target/band, 61 Vir Band 7, and three further targets are represented by their deepest single window and marked in the released catalogue. Each searched window yields its own EIRP$_{5\sigma}$ row, and Figure 1 plots only the deepest per star. When a crossing is checked against the masked species the operative discriminant is velocity coincidence, since circumstellar emission sits at rest offset by the gas's kinematics while a technosignature may appear at any frequency and drift rate.

Both carrier-lane thresholds and continuum fluxes are corrected for ALMA's primary-beam response at the star's offset from the phase centre, negligibly for most targets (median factor 1.00004, 90th percentile 1.018), though 41 of the 431 retained windows exceed 2 per cent and 31 exceed 10 per cent. The largest retained correction is j1256-1257 Band 7 at 1.81×, at $0.46\,\theta_{\rm PB}$. What decides between a retained correction and a withheld window is whether the Gaussian beam form is defensible at that offset, not the offset itself. ϵ Eri Band 6 lies beyond the first sidelobe transition, where the correction reaches 2.9–3.4× and the Gaussian form fails, so we *withhold* those 4 windows; they enter no number here and are not in the released catalogue. Under the same nominal rule they would have covered 4.1×10^{13}–2.4×10^{14} W at 3.22 pc, which at the lower end is among the deepest thresholds in the survey, and that is exactly why we decline to quote them on a beam model we cannot defend. Everything retained sits inside $0.46\,\theta_{\rm PB}$, where the correction stays a bounded systematic. The annulus geometry keeps the pipeline's own $\theta_{\rm PB} = 1.22\,\lambda/D$, an Airy first-null coefficient; recomputing them on ALMA's FWHM $1.13\,\lambda/D$ at each window's own dish diameter moves 33 of 431 windows by more than 1 per cent, extremes −7.3 to +10.3 per cent, and no headline threshold moves.

The natural benchmark here is frequency-matched and purely geometric: a 12-m aperture radiating 1 MW at 230 GHz, EIRP $\approx 8 \times 10^{14}$ W, which the median Class A effective threshold 1.7×10^{15} W misses by a factor 2.1. Thresholds are also compared to an Arecibo-like planetary radar (Drake 1974; Sheikh et al. 2025, EIRP 2.0×10^{13} W), which is an 2.38 GHz *power* scale carrying no transmitter model and is labelled in the figures as an EIRP benchmark on the power scale only: aperture gain scales as ν^2, so the same aperture and transmitter power at 230 GHz would radiate $2.0 \times 10^{13} \times (230/2.38)^2 \simeq 2 \times 10^{17}$ W. Because that comparison is between physical transmitter powers it must use $P_{\rm eff}$ and not the trigger, which carries the deepest window from below Arecibo-class power to 3.7×10^{13} W above it (§6.1).

4.4. *Ancillary and validation analyses*

Four further analyses are summarised here as far as the main text depends on them. A fifth is astrophysical and is reported with the search results: the recovery and localisation of β Pictoris CO in two transitions and four blocks (§5.3), which is the only end-to-end demonstration here that the pipeline detects a genuine narrow feature at a known stellar position.

Injection-recovery validation (Appendix B). Synthetic tones of known amplitude were injected into real visibilities and recovered with the unmodified pipeline, 1200 trials across six window configurations, three bands and both resolution classes. Two properties of the pipeline matter for reading the result. The baseline step is a per-integration block median along *frequency* (Table 2, step 4), which by construction cannot remove a feature confined to one or two channels, and the de-drift step is an inverse-variance-weighted sum over integrations, which accumulates a persistent carrier along its drift track. The withdrawn 0-of-500 coarse-window recovery was an artefact of that campaign's recovery criterion and not of the pipeline (Appendix B).

The primary statistic has no automated injection test on every pipeline change, so three quantified checks bound undetected pipeline faults instead: the machinery-only lane, whose amplitude fidelity of $-9/+14$ per cent caught the artefact above; the AU Mic re-analyses, whose recomputation agrees with the released products; and the pseudo-star rank uniformity over 220 672 pooled ranks ($p = 0.37$).

4.5. *Dwell-fraction completeness*

How long a carrier occupies one native channel defines the searched class, so we measure recovery against that directly. A carrier was injected into the retained per-integration spectra of 12 real windows, six coarse and six fine, at amplitudes of 2–20 times the per-integration noise and dwell fractions $f_{\rm dwell} = 0.10$–1.00, six realisations each: 3 888 trials, released with one row per injection (Table 3). Its detection criterion, recovered when the logged per-integration S/N reaches five, was re-derived from the released rows and agrees with the logged flag 3888 times in 3888 trials, with 0 mismatches across a clean separating gap. Recovery rises monotonically with dwell. Pooling amplitudes at or above 4σ, coarse windows recover 100 per cent of continuous carriers, 100 per cent at $f_{\rm dwell} = 0.25$ and 83 per cent at 0.10; fine windows give 100, 91 and 71 per cent.

An end-to-end check confirms this through the unmodified pipeline. Injecting a zero-drift, always-on tone into the calibrated visibilities of a coarse Band 7 window at 1, 3 and 10 times the per-integration rms returns $T_\star = 20.5$, 65.3 and 217.7, each within one channel of the injection. The ideal stacking gain over its 639 integrations is $\sqrt{639} = 25$, and the measured 20.5 is that value less weighting and flagging losses.

Two distinctions apply. The dwell campaign injects into retained per-integration spectra, after the baseline step, while the end-to-end check injects into visibilities before it; and one cell, the through-pipeline recovery of near-static carriers on Class A windows, is measured only in part.

The campaign measures completeness at the survey threshold and across configurations. The trigger acts on the stacked statistic and the de-drift step is an inverse-variance-weighted sum over integrations, so a carrier of per-integration amplitude A present for a dwell fraction $f_{\rm dwell}$ of a track of $N_{\rm int}$ integrations accumulates to

$$a \;=\; A\, f_{\rm dwell} \sqrt{N_{\rm int}} \tag{4}$$

in units of the per-integration noise. Recovery depends only on the combination a, so binning the released trials in a unifies the two axes and places all 12 injected windows on one axis. Those windows span a factor of 63 in integration count over 11 targets, two resolution classes and Bands 3, 6 and 7, so configuration-independence is measurable. In the 6 windows whose amplitude grid straddles the trigger, the 50 per cent recovery point lies at $a = 5.3$–8.3. That is the 5σ threshold itself, independent of channelisation, integration count and band. In the other 6 the weakest injection already stacks to $a = 3.4$, so they are consistent but uninformative about the curve's position. The pooled curves (Table 3) agree between classes within the sampling noise.

The injected carriers do not drift, so what is measured across 12 configurations is the persistent and partial-dwell non-drifting class. On Class B windows that is the searched class, which therefore carries a survey-threshold completeness curve; on Class A windows the searched class is the drifting carrier, whose curve is still the single calibration configuration of Appendix B, so the fine column of Table 3 bounds nothing about the transfer error. The quantity a is a derived combination, justified by the form of the statistic and by the collapse of 12 configurations onto one curve rather than measured independently at every $(A, f_{\rm dwell})$ pair.

One caveat attaches to the drift ceiling. The searched value is 12.0–13.3 Hz s^{-1} GHz^{-1}, line-of-sight $|a| = 3.60$–$4.00\,\mathrm{m\,s^{-2}}$, the generic value everywhere except the 12 windows where a known planet's own bound raises it. It clears Earth-rotation and Earth-orbit drifts by two orders of magnitude but is marginal against a close-in planet on a TRAPPIST-1 b-like orbit, so the limits of §6 are conditional on it (Figure 8).

Interference rejection, which is an operative gate. Satellite constellations grew across the 2013–2025 archival baseline, and although no ordinary downlink allocation maps into the searched frequencies, an allocation table excludes neither harmonics nor intermodulation products nor unconventional emitters, so no exclusion claim here rests on one. The

TABLE 3

COMPLETENESS MATRIX. *(a)* WHAT IS MEASURED FOR EACH SEARCHED MORPHOLOGY AND WHERE: 'E2E' IS INJECTION INTO VISIBILITIES THROUGH THE UNMODIFIED PIPELINE, 'POST' INJECTION INTO RETAINED PER-INTEGRATION SPECTRA AFTER THE BASELINE STEP, AND 'CFG' THE NUMBER OF WINDOW CONFIGURATIONS THE CELL SPANS. THE TWO CAMPAIGNS MEASURE DIFFERENT QUANTITIES: 1200 INJECTION TRIALS (APPENDIX B) AND 3 888 DWELL TRIALS OVER 12 REAL WINDOWS.

Morphology	e2e	cfg	thresh.	dwell	drift
Class A drifting	yes	1	yes[a]	no	yes
Class A near-static	post	6	no[b]	yes[e]	n/a[c]
Class B stationary	yes	6	yes[d]	yes[e]	none
Class B drifting	yes	6	yes[d]	yes[e]	none[c]
Intermittent, either class	post	12	no	yes[e]	by class

[a]42 per cent at the trigger, 83 at twice it, one configuration only (AU Mic, Band 6, 488 kHz), transfer to the other 118 Class A windows bracketed 0.5–2× (Table 7). [b]After the baseline step only. [c]Sub-channel over the track by definition. [d]Half recovery at stacked $a = 5.3$–8.3 in the 6 windows straddling the trigger. [e]83–100 per cent (coarse), 71–100 per cent (fine), over $f_{\rm dwell} = 0.10$–1.00.

(b) completeness at the trigger, against stacked amplitude a (Eq. 4), pooled over the 12 injected windows; trial counts in parentheses.

stacked a	Class B (coarse)	Class A (fine)
< 3	0% (30)	0% (72)
3–4	0% (12)	7% (42)
4–5	3% (30)	12% (42)
5–6	46% (24)	33% (48)
6–8	58% (60)	70% (96)
8–12	100% (108)	99% (174)
> 12	100% (1680)	100% (1470)

defences that do the work are the control ensemble and the time domain. A satellite enters the star and the control extractions alike within an integration; the mismatch between its bandwidth and ours, together with the drift-stacked sum over the track, dilutes a transient pass; and a second epoch at the same tuning tests whether anything reproduces. What none of that reaches is a near-field emitter, whose interferometric response need not resemble a celestial source's at all, which is one reason a per-integration RFI screen and a visibility-domain check belong in any successor analysis.

Exploratory diagnostics, which are not part of candidate selection and are collected in Appendix C. Four were run: closure phase, which has *no* discriminating power at these flux densities, and a chirped-drift extension, a periodicity search and a cross-target frequency match, all behaving as the null predicts. None derives a limit and no flagged window could have failed any of them.

What follows is the final analysis. The pipeline as released differs from earlier working versions in five substantive ways, logged with their effect on released products in Appendix H; only the one §5.4 dates changes a released number, and asymmetric-derived quantities are historical there.

Scope of the constraints. *What was searched:* 33 frequency islands totalling 93.1 GHz (88.2 GHz after molecular-line masking) at 15.3 kHz–31.25 MHz channelisation, during the analysed archival intervals, for carriers within the searched drift ceiling and one native channel in width, at nominal trigger thresholds whose measured recovery is 42 per cent, 21–84 per cent under the 0.5–2× transfer bracket (single configuration). Inside that box the non-detection is a constraint. *Nothing in this paper bounds the prevalence of:* (i) drift-rate structure on the 313 coarse windows, which carry no drift discrimination and no survey-threshold completeness curve (§4.5); (ii) transmitters emitting *outside the searched spectral windows*, meaning those 33 islands, not ALMA's frequency range (Fig. 3); (iii) transmitters active *outside the archival epochs*: the limits assume near-continuous emission and are only weakly constraining below epoch activity ∼0.1 (Table 11); (iv) drift rates *above* the searched ceiling $|a| = 3.60$–$4.00\,{\rm m\,s^{-2}}$, excluding the most close-in orbits (Fig. 8); (v) carriers placed *inside* the $\pm 50\,{\rm km\,s^{-1}}$ molecular-line mask, which is excluded search space and not searched-and-empty; (vi) emission *broader* than one native channel, for which the thresholds must be rescaled; (vii) anything below the nominal threshold, where measured recovery is 42^{+16}_{-14} per cent (Appendix B); (viii) any transmitter-frequency prior spanning ALMA's tuning range, under which the conditional bound vanishes (§6.2).

5. RESULTS

One late correction stands behind the non-detection and it made the picture more conservative: the mid-analysis statistic revision only ever removed outliers, seven windows before and four after (§5.4, Table 6).

Per-target values, including each band's searched frequency range, are in the released catalogue (Data Availability).

5.1. *Barnard's Star and Wolf 359*

Barnard's Star and Wolf 359 are the second- and fourth-nearest stellar systems. We find no prior mm or submm technosignature limit for either, so these appear to be the first. Both are non-detections in public Band 6 archival data. The nominal ${\rm EIRP}_{5\sigma}$ trigger thresholds are 6.2–6.9×10^{13} W for Barnard's Star over four windows and 7.8–9.1×10^{13} W for Wolf 359, with continuum upper limits of 0.575 and 0.453 mJy.

TABLE 4
THE SURVEY AS SEARCHED, AND THE PAPER'S HEADLINE NUMBERS IN ONE PLACE. EVERY COUNT QUOTED IN THIS PAPER RECONCILES WITH IT. THE TWO COMPLETENESS EXPERIMENTS MEASURE DIFFERENT QUANTITIES AND ARE KEPT IN SEPARATE TABLES.

Quantity	Value
Stars within 40 pc with qualifying public ALMA coverage	168
genuinely pointed at; never observed (§5.2)	115; 53
Stars searched; independent systems	88; 81
Star/band datasets	107
Spectral windows searched	431
12 m array; ACA 7 m	290; 141
extracted; repeats, defective, withheld	462; 21, 6, 4
Class A drift-resolved / Class B unresolved-excess	118 / 313 (73% Class B)
Execution blocks	102
Distance range	1.30–39.63 pc
Unique frequency union	93.1 GHz (33 intervals, 89.6–495.1 GHz)
Windows by ALMA band (3/4/5/6/7/8)	40/4/4/216/155/12
Effective after line-exclusion mask	≈88.2 GHz
Native channel widths	15.3 kHz–31.25 MHz; median 15.625 MHz
Drift-rate ceilings	1.1–5.9 $\mathrm{kHz\,s^{-1}}$ (12.0–13.3 $\mathrm{Hz\,s^{-1}\,GHz^{-1}}$)
On-source time per window	21–5274 s (median 2087 s)
Nominal $S_{\min}$	0.48 mJy–1.30 Jy (median 6.7 mJy)
Nominal trigger $P_{\rm trig}$, Class A (15.3–1953 kHz)	2.3×10^{13}–6.8×10^{16} W (median 7.5×10^{14})
Nominal trigger $P_{\rm trig}$, Class B (15.625–31.25 MHz)	1.6×10^{13}–2.8×10^{17} W (median 2.0×10^{15})
Effective $P_{\rm eff} = 2.29\, P_{\rm trig}$, Class A	5.4×10^{13}–1.6×10^{17} W (median 1.7×10^{15})
Class B	3.7×10^{13}–6.4×10^{17} W (median 4.6×10^{15})
Completeness, by class and experiment	Table 3 (amplitude); Table 3 (dwell)
Threshold crossings ($\geq 5\sigma$ at the stellar position)	20 windows (all Class A)
Star-exceeds-ring windows ($\geq 5\sigma$)	4 windows (all Class A)
circumstellar or foreground CO	3
unattributed, absent in the repeat block	1
Credible technosignatures	0

5.2. *Data-quality exclusions and open items*

107 target/bands are searched to completion and reported here. Four classes of entry are absent from the released catalogue and are itemised in Appendix H.

A further 68 candidate target/bands were selected but never delivered a searchable product, and the cause lies in the selection step. Re-running the crossmatch shows that 60 of them were never observed at all, no observing unit offered to those entries containing the star: they sit a median 1.6 degrees from the nearest field centre, because the observations were pointed at solar-system targets whose quoted field of view is enormous, and a crossmatch that trusts that quantity admits stars far outside any real primary beam. The remaining 8 entries are genuine processing failures, and re-running them on the current code returned 38 further failures in 39 attempts. A positional crossmatch must not use the quoted field of view of an ephemeris or solar delivery as a search radius; the per-star pointing ledger is released so the error can be audited. 53 of the 168 stars admitted to the candidate list fail on pointing in every band, so the sample here is 88 of at most about 115 genuinely covered stars rather than 88 of 168, and the coverage completeness of Table 4 should be read against the smaller denominator.

THE COARSE-NOISE DEFECT, AND THE BOUND ON ITS REACH

A reproducible defect produces implausibly small noise in a minority of coarse-channel windows. The radiometer relation $\sigma \propto \mathrm{SEFD}/\sqrt{N_{\rm bl}\Delta\nu_{\rm ch}t_{\rm on}}$, in the system equivalent flux density (SEFD), makes $q \equiv \sigma\sqrt{t_{\rm on}\Delta\nu_{\rm ch}} \propto \mathrm{SEFD}/\sqrt{N_{\rm bl}}$ a quantity physics predicts, so a window whose q lies far below the sample median is too extreme for a noise fluctuation and points instead to a bookkeeping defect; we exclude such windows at 100× below the median. Six of 431 windows fail, across five stars. Their nominal thresholds of 2.0–7.8×10^{12} W would otherwise have set this paper's headline depth. Excluding them, the deepest is 1.6×10^{13} W.

The exclusion does not rest on the retained population being narrow, since q runs over a factor ∼290 of the sample median, mostly the band dependence the same relation predicts through SEFD. It rests on the size of the gap and on the direction of the error. The six sit a factor 1.5×10^3 below the lowest retained window across an empty 3.2 decades, so any threshold between 1.5×10^1 and 2.2×10^4 in absolute q selects the same six, and modelling q for every retained window from its own band, array and antenna count leaves the residuals unimodal with the six 3.0 dex below the lowest retained window and nothing between. Direction settles the rest: σ is estimated from the control probes and applied to every position of a window alike, so a window-level deflation can only over-state sensitivity and can never manufacture a star-exceeds-ring window.

The six share a signature. They are all 12 m array, 3 in Band 6 and 3 in Band 7, all at 15.6 MHz, in 5 execution blocks across 4 projects, and every one has a normally behaved finer-channelised window covering the same frequencies in the same block: a channel-averaged auxiliary window riding on a frequency-division science baseband, carrying a constant WEIGHT column and so never given that baseband's calibration (Appendix H). The deflation is common mode in the data and in the weights alike, so it leaves the signal-to-noise ratio the search uses untouched, which is why the defect can misstate a threshold, cannot manufacture a flag, and costs no frequency coverage at all.

21 of the 462 extracted rows are duplicate catalogue entries for one physical window, the same (star, execution

TABLE 5

The four stage-1 spatial outliers, with the evidence dispositioning each. $T_\star$ is the search statistic at the stellar position (Eq. 2) and 'ring max' the largest of that window's 512 control statistics; $\nu_{\rm cross}$ is the crossing channel, which differs from the window centre by up to 0.85 GHz here, and $S_{\rm peak} = T_\star\sigma$ is the peak flux density in one native channel. $\Delta\nu$ and Δv give the topocentric offset of that channel from the nearest *laboratory* rest frequency of the masked species. The repaired mask of §5.3 changes no row, and the HD 48370 row is the limiting case, a near-tie of rank dispositioned by velocity coincidence with independently published emission.

Window	$\nu_{\rm cross}$ (GHz)	$T_\star$	ring max	$S_{\rm peak}$ (mJy)	$\Delta\nu$; Δv (MHz; km s^{-1})	Disposition
β Pic B3	115.260644	14.65	7.27	54	-10.56; -27.5 (CO 1→0)	circumstellar CO
β Pic B6	230.516128	11.68	9.12	78	-21.87; -28.4 (CO 2→1)	circumstellar CO
HD 48370 B6	230.511674	27.10	26.83	0.82×10^3	-26.33; -34.2 (CO 2→1)	foreground CO
CP−72 2713 B7	344.269746	5.81	5.68	12	-1526.24; -1323.2 (CO 3→2)	unattributed; no repeat

block, frequency range, channel width) key appearing twice in the archive query, so one row per key is the right statistical unit. The retention rule is deterministic and quantity-independent, the key's first-listed row in the query's own identifier order, so it consults no measured value and cannot prefer a crossing or a deeper threshold. Repeats need not agree: two reductions of the same τ Ceti Band 6 window differ by 44 per cent in threshold, which exceeds the flux-scale systematic of §4.3 and is the most direct empirical bound here on threshold reproducibility, and the two HR 1010 Band 6 rows straddle 5σ at 5.10 and 4.95, so the crossing count is rule-dependent at the level of one window. The four flagged windows are not, no discarded row being both a crossing and a stage-1 spatial outlier (Appendix H).

5.3. *Technosignature search*

All 107 star/band datasets yielded at least one carrier-lane result, 431 spectral-window measurements after the catalogue audit removed 21 duplicate rows, Bands 6 and 7 dominating (Table 9). Nominal trigger thresholds span 1.6×10^{13}–2.8×10^{17} W at a median of 1.4×10^{15} W (Figs. 3 and 1), the deepest being reached toward the nearest systems, 1.6×10^{13} W for the UV Ceti pair at 2.7 pc, though depth also tracks integration time and channelisation.

The statistics come strictly before the astrophysics, in three named stages that do not promote one another. A *crossing* is one channel×drift cell at the stellar position reaching $T \geq 5$: 20 of the 431 windows contain one. A *stage-1 spatial outlier* is a window whose stellar statistic also exceeds all 512 controls: four do. That screening device selects a window for examination (§4.2). Each of the four is then carried past that floor by a second-stage local null ((v) below), which asks whether a feature exceeds the noise at its own position without discriminating spatially; in 3 of the 4 windows the control-ring maximum clears it too, so candidacy still turns on the first-stage star-versus-control comparison. A *candidate* is a stage-1 outlier surviving every veto, and there are none. Three of the four have independent CO evidence, matched transitions, velocities and epochs at β Pictoris and published foreground emission at HD 48370; the fourth, towards CP−72 2713, never acquired one and is an unconfirmed first-block event, absent from a repeat block, below. What disposes of the three is that astrophysical evidence, with the ± 50 km s^{-1} mask playing no part. Table 5 gives each flagged window with the evidence dispositioning it.

AN OUT-OF-SAMPLE CHECK ON BLOCKS SEARCHED AFTER THE PIPELINE FROZE

The statistic and the mask were settled on these data, so a check free of that weakness needs windows the design never saw: the unsearched blocks of §3, now being processed. At the time of writing the campaign has searched 67 execution blocks toward 16 stars, 271 spectral windows in all, with the frozen pipeline and no change of any kind to statistic, mask, threshold or candidate rules after the first was searched; a further 6 blocks failed calibration and produced no search product, 1 are running, and the campaign continues. 23 of the 67 blocks are toward the 11 stars that carry no flag in the survey (GJ 849, GL 3379, HD 23484, HD 31392, HD 33793, HD 377, HD 45184, HD 53143, HR 1010, TRAPPIST-1 and Wolf 359), which is what makes this something other than a re-test of the flagged windows.

4 of the 271 windows carry β Pictoris CO at the stellar position, at $T_\star = 8.9$ to 30.8, which is what a working statistic should do where a real localised signal is present, and 9 more cross the trigger at the star inside an annulus reaching 39, which is resolved emission and not a spatial outlier. One window, toward HD 23484 (Gaia DR3 4856592239127350400) in Band 6, is a stage-1 outlier with no astrophysical attribution: a single channel at 230.790 GHz reaches $T_\star = 5.30$ against a largest control of 5.25 and a 99th percentile of 5.10, and no transition of the mask catalogue lies nearer than 252 MHz, though the wider catalogues hold one 4.6 km s^{-1} away, as they do at 19 of the 20 survey crossings. The same tuning was searched in 3 further blocks of this star and the feature does not recur: there the star reaches $T_\star = 4.14$–4.99 against control maxima of 5.49–5.67, ranking 21, 219, 381 of 512. It is the marginal single-cell event the survey's own false-alarm arithmetic predicts, at 0.37 per cent of windows against 4 of 431 in the survey itself, and the plainest demonstration we can offer that stage-1 status carries no candidate significance.

The rank distribution of the remaining 257 windows is the campaign's most consequential result, and it is adverse to the method. The star's add-one rank is not uniform. Its median is 0.44 where 0.5 is expected, 14 per cent of the ranks fall below 0.1 against 10 per cent expected and 7 per cent above 0.9 against the same, so the whole distribution is displaced towards the star rather than growing a tail. Windows inside a block share a calibration and blocks of a unit set a target, so the window-level figure ($D = 0.10$, $p = 0.01$) treats as independent what is not; clustering removes neither the shift nor its significance, 45 of the 67 blocks having a mean rank below 0.5 (sign test $p = 0.007$)

and a star-clustered bootstrap placing the median at 0.44 (95 per cent interval 0.40–0.48). Its direction is the one a small excess at the stellar position produces, which is residual continuum, deconvolution residual or position-correlated calibration error, none of which the broadband R_σ comparison can see, since it compares variances and not means. Which of them it is remains open: the displacement does not grow with channel width, as an excess proportional to flux would. The effect raises the chance of a high rank without manufacturing a spatial outlier: 0 of the 257 exceeds all 512 of its controls. It is not peculiar to the campaign either. The survey's own 431 windows lean the same way without resolving it, median 0.47 against a star-clustered 95 per cent interval of 0.42–0.51, and the two distributions are statistically indistinguishable ($D = 0.10$, $p = 0.09$), so the in-sample checks above were underpowered against a shift of this size rather than contradicted by it.

This is a *preliminary* result and we do not present it as more. It covers 16 stars in three bands, and windows within one block share a calibration and are not independent epochs. At 257 effectively null windows the two-sided Kolmogorov–Smirnov test resolves a shift of this size, rejecting $U(0, 1)$ at $D \geq 0.08$ against the observed 0.10, and a signal in a fifth of them, each in its own top 5 per cent, is caught 100 per cent of the time. On data that played no part in building it the frozen pipeline recovers the signal present and raises no candidate, and the stellar position ranks above its annulus more often than exchangeability predicts. The counts are those at this version's build date (2026 September 12, 20:17 UTC); the release will carry the campaign's state at submission.

DOES THE STELLAR POSITION CARRY ITS OWN NOISE?

The noise scale of Eq. 2 is built across the annulus, so by construction it carries no variance peculiar to the stellar position, where residual continuum, deconvolution or self-calibration residuals and position-correlated calibration error would concentrate. That is the largest untested assumption in the method, and the retained products bound it. R_σ, the star's robust residual scale over the four stage-1 windows against the median of the 8 control positions the release keeps (Appendix H), runs 0.995 to 1.001, so the star agrees with its ring to better than 0.7 per cent at 95 per cent confidence. That matters most for CP$-$72 2713. Because $T_\star$ scales inversely with the noise the star is divided by, an unmodelled broadband excess of 2.2 per cent at the stellar position would bring that window's $T_\star$ below its own ring maximum and dissolve the outlier; the bound is a factor of three smaller, so that explanation is disfavoured without being formally excluded. It is a test of variances and not of means, and it is blind to the one- or two-channel artefact the search is built to find, which the local null and recurrence address instead.

SURVEY-LEVEL FALSE-ALARM CALIBRATION

Each window-level test compares the peak signal-to-noise ratio at the star's position with the same statistic at 512 controls, and because the drift grid and the thousands of channels apply identically to both, the per-window trials factor is absorbed empirically. Under exchangeability one designated position of 513 ranks first with probability exactly 1/513, whatever the shape of the noise distribution, so a heavy tail cannot by itself produce an excess of rank-first windows. The survey expectation is 0.84 such windows and 5 are observed. Windows are not independent trials, so the reference distribution is an execution-block clustered permutation (Appendix G), which returns $P(\geq 5) = 1.7 \times 10^{-3}$, and that excess is astrophysical. Genuine celestial line emission sits at the stellar position in 3 of the 5 (Table 5), which leaves 2 against 0.84 expected. Two further calibrations agree with that reading. Only 20 of the 431 windows contain any $\geq 5\sigma$ crossing at the stellar position at all; and a prespecified Monte Carlo that promotes one control per window to pseudo-star and runs it through the operative gates, the 5σ threshold and the velocity mask, expects 0.20 flagged pseudo-target windows, in agreement with the rank budget and with the Bonferroni scale. The trials arithmetic and the residual excess it leaves are in Appendix G.

Ranking against 512 controls cannot resolve a probability below 1/513, so a second-stage local null was built to carry the four stage-1 windows past that floor. It scrambles each integration independently in channel and re-runs the identical search with the same noise map, weights and drift grid, which destroys anything coherent along a drift track while preserving each integration's own spectral autocorrelation and trials factor; its validation criteria and failure mode are in Appendix H. Three of the four windows pass validation. The two β Pictoris windows behave as positive controls, no draw of 10^5 reaching $T_\star$ in either ($p < 1 \times 10^{-5}$ and $p < 5 \times 10^{-5}$), which is what a working statistic should return where a genuine localised signal is present. For CP$-$72 2713 the null resolves the value itself, 157 draws of 10^5 equalling or exceeding $T_\star$, and an extreme-value fit to that window's own control maxima agrees to a factor 1.3; both are used with that window below. HD 48370 Band 6 fails validation outright, 73 per cent of its real controls exceeding the maximum of 10^5 null draws, because the field carries extended emission and imaging residuals that are coherent across integrations and that the scramble destroys by design, so we quote no scrambled-noise probability there. That failure is a direct consequence of retaining full dynamic spectra for only 8 of the 512 control positions, and no number quoted here rests on a parametric extrapolation of a null tail.

Exchangeability is the statistic's central assumption, and it can fail through spatially varying noise, sidelobes, correlated pixels or calibration residuals at the phase centre. For a debris-disc archive the dominant violation is none of those but resolved astrophysical emission filling the annulus, seen here in HD 48370's CO ring and in β Pictoris' recurring Band 6 window, where the ring outshines the star by 1.5×; that suppresses a star-exceeds-ring outcome instead of manufacturing one, so it costs sensitivity and nothing else; the repeat block of the *flagged* Band 6 window measures that cost (§5.3). The primary beam is settled by construction, since the statistic is formed on *uncorrected* amplitudes whose thermal noise does not depend on position and $\sigma(t, \nu)$ is one scale shared by the star and every control (§4.2), so the 513 positions are *approximately* exchangeable under a *restricted* noise-only null. Both qualifiers

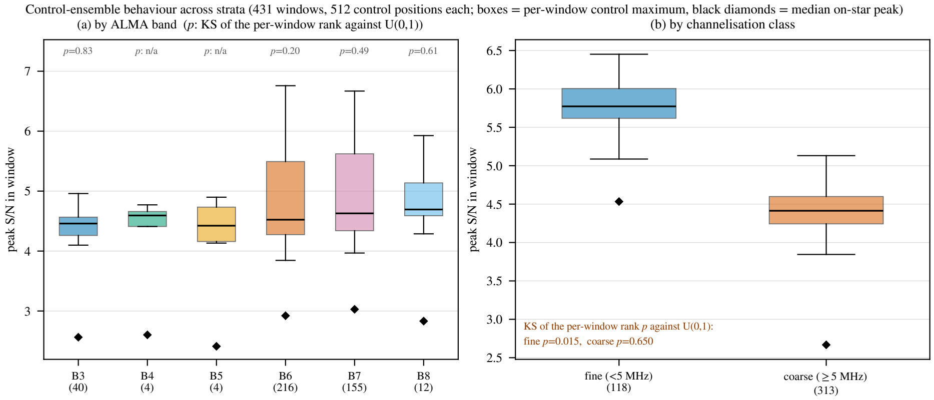

Fig. 4.— Exchangeability of the 512-position control ensemble by stratum. Boxes: per-window control maximum; diamonds: median on-star peak; p: the Kolmogorov–Smirnov test of the text. The ensemble tracks the on-star statistic in every band and both channelisation classes. The one departure is the drift-search class ($p = 0.015$), driven by the four flagged windows of Table 5.

carry weight, because that argument covers the thermal noise and the beam and does not cover residual calibration or continuum systematics, which can occur preferentially where the pointed science target sits.

Four measurements bound what is left, and all four are null. Ranking each control against the other 512-1 by the rule used for the star gives 220 672 pseudo-star ranks, uniform to the resolution the ensemble can express ($p = 0.37$) in every band and at both channelisations; that geometry cannot by itself detect a centre-versus-annulus asymmetry, so the test that can is the two-sample comparison of the 431 real stellar ranks against the pooled pseudo-ranks, and it returns $p = 0.38$. **On that comparison the stellar position behaves like one more position in its own annulus**, on these data; out of sample it does not (§5.3). The Spearman correlation of control statistic against reconstructed radius has median 0.009 over the 304 crossing-free windows, bounding any systematic radial suppression of the controls below 11 per cent in the mean. And the star's own residual variance is measured directly against the controls' in §5.3.

One stratum departs, and we report it separately. The fine-channel windows fail the $U(0, 1)$ rank test at $p = 0.015$ ($n = 118$), because four of them contain an on-star peak above their own control maximum against 0.22 expected, and those four are precisely the flagged windows of Table 5. That fits the astrophysical line emission identified below, with no misbehaving ensemble, but the departure is genuine, so fine-window rank probabilities here are slightly optimistic; excluding the 20 fine windows with any $\geq 5\sigma$ stellar crossing restores uniformity ($p = 0.39$). Two stratifications the frozen products cannot support, per-control noise records and gain, are bounded only through these empirical rank distributions. Figure 4 shows the by-stratum behaviour: median per-window control maxima are flat across bands and the fine class sits higher than the coarse, 5.77 against 4.41, as its far larger channel×drift trial count predicts.

A fully symmetric region-max variant, which maximises identically at the star and at every control, can be run on only seven locally reprocessed windows because the frozen release does not carry the products it needs. In all seven the star is exchangeable with its ring (Appendix H). The single-position calibration above remains the operative release-wide criterion, and the two are never interchanged.

ASTROPHYSICAL END-TO-END VALIDATION: β PICTORIS CO

Two of the four flagged windows are towards β Pictoris, a debris disc with resolved CO emission (Matrà et al. 2017; Dent et al. 2014): one in Band 3 and one in Band 6 (Table 5). Both sit at the same place in velocity. Carried through the frame chain, the Band 3 crossing lies $+0.44$ and the Band 6 crossing -0.29 $\mathrm{km\,s^{-1}}$ from the stellar systemic velocity, in the two lowest CO rotational transitions of a star with mapped circumstellar CO, and a third, unflagged window six years earlier shows the same component at -0.46 $\mathrm{km\,s^{-1}}$. All three lie within 0.46 $\mathrm{km\,s^{-1}}$ of systemic. We attribute both crossings to this kinematically offset CO. The full frame audit, the flux ratio and the excitation argument are in Appendix H, with the per-window numbers in Table 10.

Attributing the pair to CO does not diminish it. Against the survey background the Band 3 peak $T_\star = 14.65$ is matched or exceeded by 5 of the 431 per-window control maxima and the Band 6 peak $T_\star = 11.68$ by the same number, add-one $p = 0.014$ and 0.014. The alternative reading needs a transmitter placed by coincidence at matched offsets from two CO transitions in one star. The outliers also concentrate where the astrophysical foreground is richest: β Pic, AU Mic and CP−72 2713 are all β Pictoris moving-group members with debris discs, and HD 48370 is a disc host behind a molecular cloud (§6.3).

DISPOSITIONS OF THE REMAINING OUTLIER WINDOWS

The third stage-1 outlier, towards HD 48370 in Band 6, is the largest excess here in absolute terms and the smallest in relative ones, $T_\star = 27.10$ against a control-ring maximum of 26.83. Star and ring rising together is what emission filling the primary beam looks like. The velocities agree immediately: Cataldi et al. (2023) report prominent CO(2→1) towards HD 48370 at $v_{\mathrm{bary}} \approx 41\,\mathrm{km\,s^{-1}}$ and flag it as possible foreground cloud contamination, and our own frame chain returns $v_{\mathrm{bary}} = +41.2\,\mathrm{km\,s^{-1}}$ for this crossing, so we recover their line and confirm it kinematically. Three checks close the circumstellar alternative. The same authors report the disc itself undetected in CO and in carbon. The sight line runs through the outer Galaxy, where a flat rotation curve reaches the measured velocity only at a kinematic distance of 2.07 kpc, which nothing local could produce. And the block's ^{13}CO(2→1) window has a control maximum of 20.4 against 4.6 at the star, so the field carries bright ^{13}CO that the star does not, resolved 1.7 synthesised beams away. Appendix H gives the Band 8 [C I] tuning check, which says the same thing by omission. The rank statistic identifies nothing here, a stated limitation of the spatial test.

CP−72 2713: AN UNCONFIRMED FIRST-BLOCK EVENT, ABSENT IN THE REPEAT BLOCK

The fourth stage-1 outlier, towards CP−72 2713 in Band 7, never acquired an astrophysical attribution and was marginal throughout, its $T_\star = 5.81$ standing against a ring maximum of 5.68. Its prior for one is high, the star being a K7/M0 member of the ∼24 Myr β Pictoris moving group with a cold, dust-rich debris disc detected in ALMA 1.33-mm continuum (Moór et al. 2020), the same population and class of magnetically active star as AU Mic. We recorded the window as an unattributed stage-1 spatial outlier, a statistical or astrophysical feature and not candidate evidence, under criteria pinned in advance: *promote* on independent confirmation, *retire* on a clean second epoch or on identification with catalogued emission. That wording is committed to the released repository on 2026 September 11 (`119a68a6ab50`), 9 hours before the second block was searched.

That same member observing unit set holds a second public execution block at this tuning, `A002_Xff0235_X502d`, which the de-duplication of §3 declined. We have since calibrated and searched it with the unmodified pipeline. It matches the first block in tuning, channel width, channel count, integration count and on-source time, and the two frequency axes are registered to 0.18 of a channel, so the feature falls in the same channel. Only the drift grid differs, by one trial. The second block is also the deeper, at a combined rms of 1.964 mJy against 2.097, so a persistent emitter at the first block's flux would have appeared there at $T_\star = 6.21$. That is above both the 5σ trigger and that window's largest control, 5.93, so the pre-registered criterion had a definite prospective form: a positive recurrence meant a crossing at the first block's channel *and* drift rate reaching $T \geq 5$ in the second block and exceeding all 512 of its controls. The margin between 6.21 and 5.93 is narrow, so a second *stage-1 outlier* was never assured even for a persistent emitter. What the disposition rests on is therefore the flux comparison below, which carries no trials factor.

The feature does not recur. The first-block feature drifts. Its maximum lies at a drift of $-2.06\,\mathrm{kHz\,s^{-1}}$, a track sweeping 15.8 channels across the block, so the test belongs at that frequency *and* that drift. There the repeat block measures -1.0 ± 1.8 mJy, -0.6σ, against 11.3 ± 2.0 mJy in the first. Read as two measurements of one constant amplitude, the pair is inconsistent at 4.6σ; with the first block's flux taken as exact, which flatters the test because that flux is the largest value in the window and so biased high, the repeat block alone excludes it at 6.8σ. Maximised afresh over the repeat block's own drift grid, the same channel returns $T_\star = 3.60$ against 5.81, worth 1.6σ with an arbitrary recovered rate, and that weaker comparison is the form in which such tests are usually quoted. Two executions cannot separate a noise excursion from an intermittent emitter, and we do not claim they do.

Nothing else in the block is elevated. At the feature channel $T_\star = 3.60$ stands 0.6σ above its ten neighbours (mean 3.23, standard deviation 0.63), the noise-only expectation for a maximum over drift trials. The window's largest excursion, 4.70, ranks 188 of 513 against the same 512 controls the first block topped, add-one $p = 0.37$.

One qualification travels with this result. The two blocks are a single night's pair, 2022 October 2, the second beginning 2.05 h after the first. That makes the test strong against the class the scramble null cannot reach, since a correlator spur, a fixed-channel bandpass residual or a birdie is at its most reproducible two hours later in the same configuration and tuning, and none reproduced; and it makes the test weak against intermittency, because it constrains persistence across two hours alone and leaves the duty cycles of Appendix I untouched.

The second-stage local null replaces the 1/513 rank resolution with a measured probability, $p = 1.6 \times 10^{-3}$ from 10^5 scramble draws on a control and 2.0×10^{-3} on the star's own residual, with $p = 1.2 \times 10^{-3}$ from an extreme-value fit to this window's 512 control maxima, all three above the Bonferroni scale; the window's *ring* maximum has local-null p of 3.0×10^{-3}, only 1.9 times the star's. The scramble destroys anything coherent across integrations, and with it the instrumental classes listed above. A small local-null probability disfavours thermal noise alone and is not evidence against an instrumental origin.

Two details frame the event without bearing on its disposition. It sits in channel 35 of 3534, the most edge-proximate crossing in the survey, although edge proximity is common here, 32 of 490 recorded peak channels falling in the outer 3 per cent of their window. And $T_\star$ exceeds the ring maximum by 0.13, which is the smallest margin 512 controls can express, so that margin records a resolution floor and not a measured contrast.

Frequency disposes of the catalogue entries. A catalogued transition is stationary in the observed frame to well within a channel over two hours, while at zero drift the first-block statistic is only 2.02σ, so none can produce this feature. Re-querying the wider catalogues at the corrected 344.269746 GHz does return 3 entries in the $\pm 50\,\mathrm{km\,s^{-1}}$ tube, one of which would have masked the feature had it been in the mask of §5.3, but such proximity carries no discriminating power here: the same query at each of the 20 crossing frequencies returns a catalogued transition within $60\,\mathrm{km\,s^{-1}}$ for 19 of them.

Three further checks find nothing. The star's debris belt, resolved by Moór et al. (2020) at 140 au, lies six beams from the extraction, so the disc is not the feature; a split-half of the retained integrations gives $T_\star = 3.08$ and 5.12 against 4.11 expected for a persistent source, so both halves carry it; and the per-integration continuum of 0.07 ± 0.06 mJy would need $\epsilon \approx 70$ to drive the multiplicative bandpass route of §4.2. Polarisation hands, baseline splits, visibility-domain localisation and calibrator comparison all need re-extraction beneath the frozen products, and we claim no test we have not run. Closure phase cannot arbitrate it either: over 50 triangles the phases scatter by 92.2°, against 103.9° for phase uniform on the circle, so its mean of 35.0° ($z = 2.68$) is wrap-dominated, and at a median per-baseline

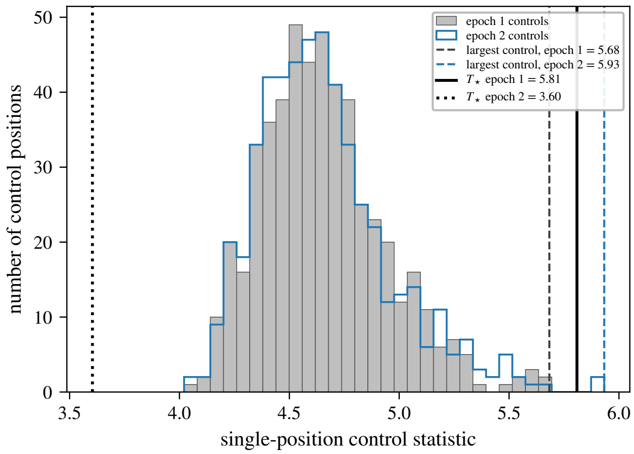

Fig. 5.— The CP−72 2713 Band 7 crossing inside its own control ensemble, in both executions. Histograms give the 512 single-position control statistics of that window in execution 1 (filled) and in the second block 2.05 h later (outline); the two agree, which is the evidence that the datasets are comparable. Vertical lines mark $T_\star$ and the largest control in each. In execution 2 the control annulus alone reaches 5.93, beyond the 5.81 that raised the flag, while the star falls to 3.60.

signal-to-noise ratio of 0.5–0.7 it is uninformative on every field run, the AT Mic binary control included (Appendix C).

The pinned criterion is met on the matched-frequency, matched-drift null of the repeat block. The standing description of this window is therefore an unconfirmed first-block threshold event that is absent in the repeat block. We find no repeat and so no evidence supporting a *persistent* technosignature at this frequency; the first-block event itself remains unexplained, it is statistically unexceptional at the survey level, and its absence two hours later does not identify it as noise. An intermittent emitter is not excluded by two executions, and we do not say it is.

A non-recurrence is only meaningful if the same machinery can recover a repeat when one exists, so we applied it to emission we already believe in. The unit behind the Band 3 flag holds 4 blocks and all 4 are now searched unmodified, an exhaustive control. The CO$(1 \to 0)$ feature is present in every one, at $T_\star = 14.65, 17.29, 27.68, 30.76$, with 17, 24, 24, 24 channels above threshold and the control maximum below the star throughout. Those executions share one scheduling block within 23 h, so they are one epoch repeated. A second Band 6 CO$(2 \to 1)$ window at 230.528134 GHz repeats the test in another exhaustive unit, 2 of 2 blocks searched. It crosses the threshold in both without becoming a stage-1 spatial outlier, at $T_\star = 10.16, 9.95$ on an exact configuration match. The integrated line fluxes agree to -0.7σ. It is Class A, so the recurrence is demonstrated at the survey's finest drift discrimination. The stage-1 Band 6 window at 230.516128 GHz has since been tested on the same terms. Its unit's second block, A002_Xd9668b_Xa9df, was calibrated and searched with the unmodified frozen pipeline on 2026 September 12 in the configuration of the first, 385 integrations of 3,534 channels at 15.26 kHz, and the feature recurs. The star reaches $T_\star = 8.95$ over 106 channels above threshold, 102.5 kHz from the first block's peak, and that separation is 0.13 km s^{-1} against a line 8.9–10.2 km s^{-1} wide. Every β Pictoris CO feature this survey flagged has now been recovered in an independent execution block, and CP−72 2713's has not.

The spatial ranking carries a warning with it, and we state it plainly. The repeat block is again a stage-1 outlier, none of the 512 controls reaching the star, but the margin has collapsed, from 1.28× its largest control to 1.06× as the star falls from 11.68 to 8.95. The controls carry the diagnosis: their maxima, 9.12 and 8.41, stand against 99th percentiles of 6.11 and 5.99 in the two blocks, so in each a few control positions hold the belt while the rest hold noise. Pinned to the first block's own peak frequency instead of maximised over the window, the star returns 8.10, and 2 controls exceed that. The cause is the one already identified for the neighbouring Band 6 window: the CO belt fills the control annulus, so the controls rise with the star. **The spatial screen loses power against emission extended on the scale of the annulus.** That is a property of the method rather than of this line, and it is why a recurrence test at a known frequency is the stronger instrument.

The recurring window also locates the emission. In both blocks the annulus is brighter than the star (ring maxima 15.17, 15.08 against $T_\star = 10.16, 9.95$, 8 and 6 of the 512 controls above the star), and that annulus, 3.82–21.27 arcsec, sits on this disc's CO belt, peaking near 85 au = 4.3 arcsec at 19.6 pc (Matrà et al. 2017). The window is therefore not a stage-1 outlier because the source fills the ring, and the frozen products place it off the star, though only coarsely: the window's 4.29-arcsec synthesised beam exceeds $r_{\rm in}$, so the innermost controls are not resolved from it.

Frequencies are compared after removing each block's own tuning, which ALMA sets per date, absorbing most of the barycentric difference; on that basis the Band 3 peaks agree to 1.00 channel and the Band 6 peaks to 3.00 channels. In the barycentric frame, and now in the ordinary radial-velocity convention rather than the frequency-offset sign of Table 10, the CO peaks of the 4 blocks with a recorded epoch span +19.6, +20.2, +20.4, +20.7 km s^{-1}, within 0.7 km s^{-1} of the adopted systemic velocity across two transitions and 8.4 yr, 2013-10-06 to 2022-03-03. Widths place the same conclusion beyond a frequency coincidence: the channel counts above threshold correspond to 10.8–15.2 km s^{-1} in Band 3 and 8.9–10.2 km s^{-1} in Band 6, both an order of magnitude above the one channel a carrier occupies. The β Pictoris test establishes that the machinery preserves a real repeat; what establishes that *this* repeat would have been seen is the $T_\star = 6.21$ expectation above.

TABLE 6
EVERY WINDOW FLAGGED BY THE ORIGINAL REGION-MAX STATISTIC, UNDER BOTH STATISTICS, SO THE REVISION CAN BE AUDITED ACROSS ALL OF THEM, NOT JUST THE AU MIC WINDOW. $S_{\rm reg}$: REGION MAXIMUM OF EQ. 2 OVER $n_{\rm src} = 135$ POSITIONS; $T_\star$: THE SYMMETRIC SINGLE-POSITION STATISTIC; 'CTRL MAX', 'RING MAX': THE LARGEST CONTROL STATISTIC UNDER EACH. THE ν COLUMN IS EACH WINDOW'S *centre*; THE CROSSING FREQUENCY DIFFERS BY UP TO 0.85 GHz, AND TABLE 5 CARRIES $\nu_{\rm cross}$. FOR CP−72 2713 THE REGION MAXIMUM ALREADY LAY ON THE STAR, SO THE TWO COINCIDE.

			region-max statistic		symmetric statistic		
Star	Band	ν (GHz)	$S_{\rm reg}$	ctrl max	$T_\star$	ring max	Symmetric outcome
HD 48370	6	230.7305	27.95	26.83	27.10	26.83	flagged
β Pic	3	115.2605	20.31	7.27	14.65	7.27	flagged
β Pic	6	230.5164	12.78	9.12	11.68	9.12	flagged
AU Mic	6	230.5384	5.97	5.85	5.22	5.85	not flagged
ALMA J1537−3319	6	217.0183	5.83	5.63	5.09	5.63	not flagged
CP−72 2713	7	345.1152	5.81	5.68	5.81	5.68	flagged
HD 14055	7	345.1377	5.80	5.74	5.22	5.74	not flagged

LINE EXCLUSION: DEFINITION, REPAIR, AND COST

The exclusion regions are defined before any crossing is classified, and they are *excluded parameter space*, not a veto: the experiment declines to classify those frequencies rather than declaring anything found in them astrophysical. The searches as executed used an automated line check matching crossings to catalogued rest frequencies within one spectral channel at zero drift. The β Pictoris windows showed that inadequate: both fell outside its one-channel tolerance, by 10.6 and 21.9 MHz, while lying, after the frame chain, within $3.6\,{\rm km\,s^{-1}}$ of the stellar systemic of a resolved, mapped disc, the regime a channel-width tolerance is blind to. Every crossing was therefore reclassified with a velocity-space mask, $\pm 50\,{\rm km\,s^{-1}}$ in the stellar frame around each catalogued transition. Its kinematic justification, empirical conservatism and the counter-argument that a communicator might deliberately choose these natural reference frequencies are in Appendix E. The mask is applied *retrospectively*, as a post-hoc classification criterion and never a detection threshold. The half-width was adopted after the β Pictoris crossings had been seen. A sensitivity test reclassifies all 20 crossings at ±20, ±30, ±50 and $\pm 100\,{\rm km\,s^{-1}}$, costing 2.1 to 10 per cent of gross bandwidth. No width admits a new unattributed outlier, and at every one of them CP−72 2713 is the only unmasked one. Masked regions are reported throughout as *excluded search space*, not as searched-and-empty.

The mask as executed carried four defects, repaired here and itemised in Appendix H. The repaired catalogue holds 17 transitions of 11 species at full JPL/CDMS precision (Pickett et al. 1998; Müller et al. 2005), applied in the LSR frame as well as the stellar frame and re-applied to the frozen crossing list as a post-hoc filter. **No disposition changes.** The cost of masking is measured on one definition, every JPL catalogued transition of the 9 frozen species displaced by each system's radial velocity, given a $\pm 50\,{\rm km\,s^{-1}}$ stellar-frame half-width, intersected with each window and merged: 5.3 per cent of the unique-frequency union and 3.1 per cent of the window-summed gross bandwidth, with CN and CO dominating the loss. Total exposure is $1.4 \times 10^6\,{\rm s\,GHz}$, or 393 star-hour-GHz, with 30 per cent of it in the five best-populated 2-GHz bins, which any frequency-prior calculation must weight (Appendix D). The mask governs dispositions and leaves the noise-derived thresholds alone.

Masking is a design selection effect. A signal coincident with a molecular line is not thereby "false"; it is *excluded from the technosignature search domain*, because the astrophysical prior there is overwhelming and this pipeline cannot separate the two. We place no constraint on transmitters in the excluded regions, 5.3 per cent of the unique-frequency union, and the release keeps those channels for the second of the two searches this dataset supports: a clean-frequency search, in which line coincidence is a veto, and a line-coincident search, in which a crossing is admitted only on independent evidence of spatial localisation, line morphology, polarisation, recurrence or non-thermal width. This paper reports the first; the β Pictoris crossings are what the second would have to handle. The exposure is not confined to the flagged windows either: 131 of 431 windows contain an in-band catalogued transition of the masked species, and in none of them does a crossing fall within one native channel of a rest frequency.

5.4. *The superseded statistic*

An earlier region-max statistic maximised over a region centred on the star against single-position controls, which inflates the star's rank. The analysis here uses the symmetric single-position statistic of Eq. 2 throughout, all 431 windows were reassessed under it, and Table 6 audits every window the original flagged: the symmetric criterion flags a strict *subset*, seven windows before and four after, so no result here depends on the superseded version. The three that drop out, among them a crossing towards AU Mic, do so because the stellar position never exceeded its own control ensemble. The redesign postdates the AU Mic flag by 9 days and we do not claim otherwise; Appendix H gives the chronology with its commit hashes, the algebra that owes nothing to these data, and the three tests AU Mic fails. For the one flagged window that survives, the order is the other way round: CP−72 2713 first appears as a flagged window in the very commit that adopts the symmetric statistic (0c465f2661bd, 2026 September 9), and its region maximum already lay on the star, so both statistics return the same $T_\star$ and no version of the analysis could have created it.

62 stars hold further public blocks this survey did not take (§3), which §6 turns into an epoch-activity constraint.

5.5. *Ancillary results*

Three by-products of the search are reported for completeness. None of them carries a technosignature disposition.

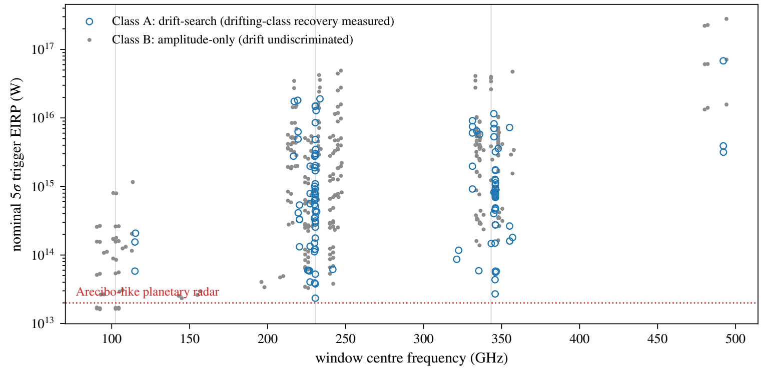

Fig. 6.— Two-dimensional sensitivity: nominal EIRP trigger threshold (boxed rule, §4) against window centre frequency for all 431 searched windows, symbol by channelisation class, the axis along which completeness differs (Table 3). Ringed symbols are the 9 windows whose intra-integration smearing correction exceeds 1 per cent. Grey lines mark the Bands 3, 6 and 7 concentrations; dotted, the Arecibo 2.38-GHz EIRP benchmark, a power scale only.

Integration-time audit (machine-readable release). Two of the 58 star–execution-block groups over-count their on-source time against the archive's own exposure accounting, AT Mic by a factor 1.43 and CD−38 10980 by 7 per cent, the rest being conservative; no EIRP threshold depends on integration time, so exposure-weighted coverage is optimistic by under 1 per cent (Appendix G).

Continuum search (Appendix F). Of the 57 target/bands processed far enough to measure, seventeen give $\geq 5\sigma$ detections and forty non-detections reaching 0.036–3.6 mJy (median 0.52 mJy). Every detection is astrophysically unsurprising, the auroral emitter LSR J1835+3259 meriting deeper follow-up.

Serendipitous molecular-line catalogue (machine-readable release). Across the subset scanned, twelve $\geq 5\sigma$ outlier groups are single- or few-channel features with no *masked* transition within 300 MHz and six are plausibly real, all reported openly. Proximity is measured against the 17-transition mask of §5.3, never the wider catalogues.

6. DISCUSSION

6.1. *What signals could ALMA archival data actually detect?*

Each morphology is illustrated with a real window, the completeness entries of Table 3 and Fig. 6. Denominators count the 81 systems whose deepest threshold in the class lies at or below the case power.

(i) A continuous, frequency-stationary carrier in a coarse channel is the best case. The UV Ceti pair at 2.7 pc has a deepest window whose trigger is 1.6×10^{13} W and whose effective threshold is 3.7×10^{13} W, and a channel-confined carrier there is recovered at near-ideal $\sqrt{N}$ gain (§4.5). Read on the trigger, 1 system of 81 reaches Arecibo-class total power and 2 reach twice it; read on the effective threshold, which is the comparison a transmitter power calls for, 0 do and 1 does. The survey's best case is therefore a few times Arecibo-class power.

(ii) A continuously drifting carrier in a fine 488-kHz channel is the only morphology with a measured end-to-end recovery curve (Appendix B): above $\sim 8 \times 10^{13}$ W nominal it is recovered in 83 per cent of trials with its drift inside the searched ceiling, 42–100 per cent once the 0.5–2× transfer bracket is carried, and 8 systems reach that depth in Class A.

(iii) An intermittent carrier, on for a quarter of the track, above $\sim 2 \times 10^{14}$ W is detected toward 24 mid-distance systems. Recovery falls rapidly at smaller dwell fractions (Table 3): below $f_{\rm dwell} \sim 0.1$ only the nearest windows carry information.

6.2. *What does the null result mean?*

No credible technosignature has been found (§5.3). The $\mathrm{EIRP}_{5\sigma}$ values are nominal trigger powers (§4), and completeness-calibrated exclusion holds only in Class A, where the measured recovery curve applies.

One exclusion deserves the converse emphasis. The line mask removes $\pm 50\,\mathrm{km\,s^{-1}}$ stellar-frame tubes around the catalogued transitions, whose rest frequencies are the 'magic' channels Cocconi & Morrison (1959) argued a communicator might choose. This survey is blind by construction to a transmitter parked there in the stellar frame, a false-positive control purchased with 3.1 per cent of gross bandwidth and one physically motivated class of signal. A carrier deliberately placed on CO(2→1) is not undetectable by ALMA, it is unsearched here, and those channels are retained in the release for anyone bringing independent spatial or time-domain discrimination. Of the 41 planets known around the 18 hosts, only TRAPPIST-1 b reaches the drift ceiling in its edge-on worst case, on accelerations $GM_\star/r^2$ from NASA Exoplanet Archive parameters maximised over phase and inclination. Figure 8 shows the ceiling as a selection function of period and host mass, the host-mass dependence being weak ($a \propto M_\star^{1/3}$ at fixed period). Coverage eligibility and search completeness differ: §3 audits the 448 public blocks in scope against the 102 searched, and the campaign of §5.3 is working through the remainder.

The survey sits against Wright et al. (2018)'s haystack axis by axis. On frequency only Band 3 falls inside that axis's 10 MHz–115 GHz span, 20.6 of 115 GHz, and the remaining 72.5 GHz (Bands 4–8) lies wholly outside it and is new territory. On sensitivity the median threshold detects a 10^{13} W transmitter only within ∼0.9 pc. Against the sole prior ALMA-archival search (Mason et al. 2024, 28 bycatch targets at ≥1.01 kpc), the 81 systems here reach median per-target thresholds ∼1250× deeper. We do not combine these fractions into one coverage number, because the axes are strongly coupled and a product of marginals would misstate the overlap by orders of magnitude. The conditional searched-domain fraction of Appendix I is that figure of merit conditioned rather than marginalised, and is not interchangeable with the 6×10^{-18} above.

What the zero detection bounds is conditional on each target's searched frequencies, epochs and morphology class, and turning it into a fraction of systems needs a per-system detection probability, which is not a demographic occurrence rate. We call it the *conditional searched-domain transmitter fraction* throughout, and Appendix I gives its framework as an *illustrative calculation only.* **It is not an occurrence rate and should not be quoted as one.** It is conditional on this survey's frequencies, epochs, duty cycles and morphology class, on a sample selected by ALMA's own proposal history, and on a transmitter-frequency prior confined to the searched islands. The result of this paper is the search sensitivity, the exposure and the null.

In plain terms, ALMA has observed 88 nearby stars as a by-product of other science, across parts of the millimetre band no technosignature search had examined. We looked at each for a channel-confined spectral carrier, and found none the surrounding sky could not also produce. The constraint is conditional on the few gigahertz, the few hours, the single morphology class and the duty cycles each star happened to be observed with, and on those terms a non-detection says little about how common transmitters are.

6.3. *What this sample can and cannot speak for*

Selection is on archival coverage, and ALMA's archive was not built to sample stars. 79 of the 88 searched stars were observed under proposals categorised *Disks and planet formation*, and 73 carry the *Debris disks* science keyword (67 of 81 systems). Among the stars with a Gaia temperature, 41 per cent are M dwarfs against 69 per cent of the reference census (Table 8), so the sample is F- and G-enriched and M-dwarf deficient. Window counts are concentrated too: three systems carry 9.0 per cent of the searched windows.

Two consequences follow, and they run in opposite directions. A debris-disc-dominated sample is the worst case for astrophysical false positives at millimetre wavelengths, which is where the stage-1 outliers of §5.3 concentrate, so the vetting burden here is heavier than a blind sample would carry. And because M dwarfs host most of the habitable-zone planets in the solar neighbourhood, **this null result carries very little weight for the habitable-zone or Sun-like-star question.** It constrains transmitters around the stars ALMA happened to point at, not the local stellar population.

Useful subsamples still fall out of archival selection. 18 of the 88 processed stars (41 known planets) are confirmed exoplanet hosts, among them Proxima Centauri, τ Ceti, ϵ Eridani, β Pictoris and the seven-planet TRAPPIST-1 system, while HN Lib and LHS 1140, each with a habitable-zone planet, now carry an ALMA technosignature non-detection. Four searched stars are white dwarfs (Gentile Fusillo et al. 2021), all continuum non-detections and among the few mm/submm SETI constraints for that class (Huang et al. 2026).

6.4. *A next-generation ALMA technosignature search*

ALMA's advantages here are its mm-wave sensitivity, an essentially unsearched tuning range, the interferometric spatial discrimination that supplies the control statistic of §4.2, and a public archive re-processable without new allocations. Six changes would enlarge what this experiment can say, ranked by the signal space each buys.

Search the blocks already held. 346 of the 448 public execution blocks in scope were never downloaded (§3), and searching them costs no telescope time. The campaign of §5.3 has begun that work and continues; completing it would make a multi-epoch survey of 62 stars. One caution carries over from the worked case, whose two blocks are repeat executions of a single scheduling block on one night: a second block often supplies a baseline of only hours, and which it is can be settled with one archive query before any download.

Use the visibilities. The largest unused advantage is the interferometric phase itself, which says whether a spectral feature has the signature of a celestial point source at a known position. Comparing the likelihoods of a source at the propagated stellar position, a source elsewhere, common-mode interference and thermal noise is strictly more powerful than ranking a peak against a control ensemble, and it would demote the ensemble to a calibration layer.

Drift discrimination on coarse windows. Within the searched ceiling every in-grid drift on a 15.625-MHz window is sub-channel over a track, so 313 of 431 windows measure amplitude but cannot distinguish a drifting carrier from a stationary one (§4.5). Finer channelisation, or a drift model spanning several blocks, would recover that axis.

Retain more control spectra. Full dynamic spectra survive for only 8 of the 512 control positions, which is what breaks the second-stage null in a structured field and what limits the variance test of §5.3. Retaining all of them costs storage and nothing else.

Extend the injection campaign and the polarisation axis. The completeness curve should be measured on a stratified grid over band, channel width, integration time, antenna count and primary-beam position, which are the axes Table 7 shows the present single-configuration measurement does not span, and pushed past 10σ until it asymptotes. The polarisation axis buys discrimination against weakly polarised astrophysical backgrounds at no sensitivity cost.

A boundary on drift linearity. The search assumes linear drift over a track, and curvature matters once $\frac{1}{2}\ddot{\nu}T^2$ exceeds half a native channel, which only the shortest-period orbits reach: $P \lesssim 1.6$ d for a typical Class A window (230 GHz, 488 kHz, $T \approx 2000$ s). TRAPPIST-1 b at 1.51 d sits just inside that boundary and is also the case that reaches the drift ceiling.

7. CONCLUSIONS

What was searched. The public ALMA archive already holds a technosignature search of 88 stars in 81 systems within 40 pc: 107 target/band datasets, 431 spectral windows and 93.1 GHz of unique sky frequency between 89.6 and 495.1 GHz. The sample is archive-target-complete but epoch-incomplete. Every star the public archive covers is searched, in 22.8 per cent of the execution blocks those targets hold, and the stars themselves are 0.50 per cent of the catalogued population in that volume. Channelisation splits the search in two, 118 windows of drift-resolved carrier search against 313 of unresolved spectral excess (§4.2).

What was found. Nothing, in the domain searched. The $\pm 50\,\mathrm{km\,s^{-1}}$ molecular-line mask is part of that qualification: it removes 3.1 per cent of gross bandwidth around transitions whose rest frequencies are exactly the channels a communicator might choose, and those channels are unsearched here because the analysis vetoes them, while ALMA can see them perfectly well. Four windows are stage-1 spatial outliers, and three of those are CO emission: two circumstellar components towards β Pictoris and one previously reported foreground cloud towards HD 48370 (Cataldi et al. 2023). The fourth, towards CP−72 2713, never acquired an astrophysical attribution. A second public execution block at the same tuning, searched after the pre-registered criterion had been pinned in the released repository, shows no repeat at the feature's own frequency and drift. We therefore find no evidence supporting a *persistent* technosignature, and report 0 credible ones; two executions cannot separate a noise excursion from an intermittent emitter (§5.3). Barnard's Star and Wolf 359 carry their first published mm/submm limits (§5.1).

The defensible sensitivity and search domain. The nominal $\mathrm{EIRP}_{5\sigma}$ values are trigger powers and not completeness limits (boxed rule, §4). They run 2.3×10^{13}–6.8×10^{16} W over Class A and 1.6×10^{13}–2.8×10^{17} W over Class B. The number to set against a transmitter is the effective unresolved-carrier threshold, $\times 2.29$ higher under the instrument's spectral response, 5.4×10^{13}–1.6×10^{17} and 3.7×10^{13}–6.4×10^{17} W, and on that scale 0 of the 81 systems reach the total power of an Arecibo-like planetary radar. Neither figure is a completeness limit: recovery is calibrated end to end on one configuration (Appendix B) and transfers to the others only within a bracket of 0.5–2×, so what the survey holds is a reference recovery function whose transferability is incompletely established. The conditional searched-domain transmitter fraction stays illustrative, and spans 5.0–12.3 per cent across that bracket alone (Appendix I), and the *Scope of the constraints* box of §4 fixes what the non-detection binds.

The implication for future ALMA archival SETI. 346 of the 448 public execution blocks in scope had never been searched when this analysis was frozen. They need no allocation and no new observing time, and they would turn much of this single-epoch experiment into a multi-epoch one over 62 stars. That work is under way: 67 blocks toward 16 stars, 271 windows, have been through the frozen pipeline since. They recover the one real signal among them and raise one unattributed stage-1 outlier, the rate this survey's own false-alarm arithmetic predicts; but the stellar position ranks above its annulus more often than exchangeability predicts (median 0.44 against 0.5), so the campaign confirms the false-alarm rate and falsifies uniformity, and the spatial screen is mildly anti-conservative (§5.3). The campaign continues, and §6.4 ranks the other extensions.

ACKNOWLEDGEMENTS

This paper uses ALMA data from the public ALMA Science Archive. The 102 execution blocks searched, with project codes and epochs, are listed in the machine-readable release and should be cited from it in the standard JAO.ALMA#XXXX.X.NNNNN.S form. They span 2013 October 6 to 2025 June 11 across 36 proposal codes. The two blocks behind the AU Mic recurrence test of §5.4 are from project 2012.1.00198.S, executed 2014 August 18 and 2014 June 5. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), MOST and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (CDS, Strasbourg, France).

AUTHOR CONTRIBUTIONS (CREDIT)

Glenn J. White: Conceptualisation, Methodology, Supervision, Writing – review & editing.
Robin Dey: Software, Investigation, Formal analysis, Data curation, Visualisation, Writing – original draft.

DATA AVAILABILITY

The reduced data products underlying this paper, with the full target-selection and analysis pipeline, are at `https://github.com/Tilanthi/SETI`. All numerical values here come from the *submitted-analysis snapshot*, the repository commit tagged at submission, which freezes pipeline code and input data as run and reproduces every number, table and figure from the regeneration scripts shipped with it.

The machine-readable catalogue, `per_target_results_v3.52.csv`, carries 431 rows, one per searched window, in 41 columns. Each row gives star name and machine-readable `system_id` (the independent stellar system that is the statistical unit, seven bound pairs sharing one identifier), distance, ALMA band and execution block, searched frequency range, native channel width, on-source time and integration count, per-channel rms, the nominal $\mathrm{EIRP}_{5\sigma}$ trigger with both response-corrected values (§4), the drift ceiling as a rate and as an acceleration, the trial drift count, η_{drift}, η_{smear}, resolution and search class, stellar and control peak statistics with the count of controls at or above the star and the add-one rank probability, the crossing frequency, the nearest catalogued transition with its offset, the full control geometry (ring centre, θ_{PB}, r_{in}, r_{out}, control count, seed) and the disposition, verbatim from Table 5. The blocks searched by the epoch-extension campaign of §5.3 lie outside this frozen release, and their per-block statistics ship in the accompanying epoch-extension record.

The complete release will also be deposited on Zenodo, with a DOI minted on final acceptance and inserted here. It carries the per-window search products for all 431 windows, including the target statistic with all 512 control statistics, from which every rank and p-value here is recomputable; the line-exclusion mask in both its frozen and repaired forms; the execution-block metadata and per-star pointing ledger of §3; the 1200 injection and 3 888 dwell trials; the second-stage local-null code with its per-window outputs; the repeat-block products behind the CP−72 2713 disposition; and the frozen pipeline code and input-data snapshot with its change-log, in which the promote and retire criteria are pre-registered, so the commit hash and timestamp quoted in §5.3 are publicly verifiable.

 Raw ALMA visibility data are public from the ALMA Science Archive at `https://almascience.org`.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; calibration, imaging and the local reprocessing of §§5.3–5.4), Python 3 with astropy, numpy and scipy (spectral extraction and statistics), and the ALMA Science Archive and SIMBAD TAP interfaces; all are free and open-source, with exact versions recorded in the frozen release.

APPENDIX

A. FREQUENCY CONVENTIONS, LINEWIDTH READING, AND SMEARING

Reference frames. The line mask's chain runs topocentric → barycentric (Earth ephemeris at the execution-block mid-time) → stellar rest frame (catalogued systemic radial velocity), with JPL/CDMS rest frequencies (Pickett et al. 1998; Müller et al. 2005); in the LSR frame it stops at the standard solar motion, not the systemic velocity (§5.3). The residual error is the catalogue radial-velocity uncertainty alone, a few $\mathrm{km\,s^{-1}}$ within 40 pc; second-order relativistic terms at the $50\,\mathrm{km\,s^{-1}}$ scale are $\lesssim 5$ kHz at 350 GHz, below the finest native channel searched (15.3 kHz). The β Pictoris and AU Mic attributions use the same chain, so their offsets are comparable to the exclusion tolerance.

Channel widths, and the instrumental response. These come from the measurement sets' own spectral window tables (CHAN_WIDTH). Earlier versions read a missing "EFFECTIVE_BANDWIDTH" column as evidence of no online spectral averaging; MS v2 defines no such column, so that check is withdrawn, and with it the claim that the quoted channelisations are noise bandwidths.

ALMA applies online Hanning smoothing by default, so the spectral response is ∼2 channels wide and neighbouring channels are correlated. For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, with $\sigma_{\rm smoothed}/\sigma_{\rm raw} = 0.612$: the correlation is not confined to adjacent channels, the lag-two term being a sixth. A sub-channel tone keeps 0.50 of its power in the peak channel at the channel centre and 0.38 on a boundary, median 0.44 over uniform placement. Every $\mathrm{EIRP}_{5\sigma}$ here is therefore optimistic by ×2.00 to ×2.67, median ×2.29; the released catalogue carries the nominal threshold and both corrected values per window, and the sensitivity summary quotes the worst case. The injections deposit an unsmoothed delta-in-channel tone, so the recovery curves of §4.5 and Appendix B carry the same bias and the end-to-end check (Eq. 4) cannot see it. The retained per-integration residuals of the second-stage local null (§5) have a measured lag-one spectral autocorrelation of 0.65, against the Hanning prediction 0.667, so that correlation is instrumental and the residual spectra are smoothed and un-averaged, which is the independent confirmation of the smoothing state. The archive's reported effective spectral resolution per window, against each measurement set's own channel separation, is exactly 2.00 in 403 of the 431 windows, in both correlator modes and including 309 of the 313 coarse time-division-mode (TDM) ones; the other 28 report a smaller ratio, the signature of online channel averaging, where the peak-channel loss is smaller. The blanket factor is right for the great majority of both classes. We still do not rescale, so each threshold here is nominal and up to ×2.67 optimistic.

Linewidth convention. The statistic is the single-channel flux density after drift correction, so the quoted $\mathrm{EIRP}_{5\sigma}$ applies to any transmitter narrower than the native channel and is flat in the assumed linewidth. Two sub-channel qualifications attach. Flatness is exact only for a centred tone, a boundary-straddling one splitting its power between two channels at worst by a factor of two. And the channel-to-channel transition follows the correlator's passband shape, which the frozen products do not retain. Thresholds therefore carry a sub-channel-position systematic of tens of per cent, and EIRP values are quoted to two significant figures.

Transmitters wider than one channel. The baseline filter (§4.2) is flat for features narrower than roughly half its effective width W and falls to zero above W, so the survey's sensitivity is to channel-confined morphologies and W is the linewidth ceiling: 65–74 channels over the fine class and 30–59 over the coarse one, that is ≈32 MHz at 488 kHz and ≈0.9 GHz at 15.62 MHz. Within that ceiling, a top-hat transmitter spanning $N_{\rm ch}$ native channels has per-channel flux the total divided by $N_{\rm ch}$ at unchanged noise, so to first order

$$\mathrm{EIRP}_{5\sigma}(N_{\rm ch}) \;\simeq\; N_{\rm ch} \times \mathrm{EIRP}_{5\sigma}(1), \qquad N_{\rm ch} \lesssim W/2, \tag{A1}$$

the rescaling the main text applies to signals broader than one channel.

Worked example: from flux to EIRP. Take UV Ceti (G 272-61B) Band 3, the deepest of its 4 windows in that band (execution block A002_X877e41_X14c1, lower edge 91.565 GHz). Its per-channel rms is 0.241 mJy, so $S_{\rm min} = 5 \times 0.241 = 1.21$ mJy in one 15.625-MHz native channel, or $1.21 \times 10^{-29}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$ at 2.67 pc, and $\mathrm{EIRP}_{5\sigma} = 4\pi d^2 S_{\rm min}\Delta\nu = 1.6 \times 10^{13}$ W.

Intra-integration smearing. The frequency excursion at the maximum searched drift rate across one integration, $\dot\nu_{\rm max} t_{\rm int}$, is at most 1.10 channels across the 431 windows (median 0.001, 90th percentile 0.06). A linear drift smeared

TABLE 7

How far the 118 drift-resolved (Class A) windows sit from the single configuration on which recovery was measured: AU Mic, Band 6, 488 kHz channels, 1506 s on source, 49 drift trials, star on the phase centre. A window counts as matching an axis if it lies within a factor of two of the calibration value, or within $0.2\,\theta_{\rm PB}$ of the phase centre. Antenna count is not recorded in the frozen products, so it is not an axis here; the primary-beam width carries part of that information.

Configuration axis	Calibration value	Windows matching
Band	6	63 of 118
Channel width	488 kHz	101 of 118
On-source time	1506 s	98 of 118
Drift trials	49	63 of 118
Offset from phase centre	0	106 of 118
Axes on which a window differs: none		32
one		26
two		47
three or more		13

by Δ channels within an integration keeps a fraction $\eta_{\rm smear} = {\rm sinc}(\pi\Delta/2)$ of the peak-channel amplitude, so the effective threshold is ${\rm EIRP}_{5\sigma}/\eta_{\rm smear}$. The 9 windows with $\eta_{\rm smear} < 0.99$, as star, band, ($\eta_{\rm smear}$, threshold multiplier), are HIP 10679 Band 6 (0.574, ×1.74), β Pic Band 6 (0.574, ×1.74), HD 172555 Band 6 (0.574, ×1.74), HIP 10679 Band 6 (0.606, ×1.65), HD 172555 Band 6 (0.606, ×1.65), η Crv Band 8 (0.865, ×1.16), η Crv Band 7 (0.932, ×1.07), HD 48370 Band 6 (0.979, ×1.02) and CP−72 2713 Band 6 (0.979, ×1.02). The remaining 422 agree with their nominal thresholds to better than 1 per cent. Compounding the worst smearing penalty with the worst Hanning factor gives ×4.6 on the nominal threshold for those five windows, and no released column carries that product. The ${\rm sinc}(\pi\Delta/2)$ form is the conservative reading: a pure occupancy argument would retain $\min(1, 1/\Delta)$, giving 1.0 rather than 0.64 at $\Delta = 1$.

B. INJECTION-RECOVERY SENSITIVITY VALIDATION

To test whether a 5σ threshold behaves as idealised Gaussian noise predicts, we inject synthetic tones of known amplitude into real visibilities and measure what the unmodified pipeline returns. The injection uses the extraction code's own mechanism, as far upstream of the statistic as the retained products allow. Six configurations span three bands and both classes: three coarse Band 7 spectral windows of AT Mic, one coarse window each in Bands 3 and 6 of AU Mic, and AU Mic's 488-kHz flagband window with its 49-trial drift grid. Each receives a complex tone at the star position at amplitudes 4–10× the statistic's own noise. The grid covers drift fractions $f_{\rm drift} = 0$ to 1.0 of the ceiling, on- and half-grid rates, and channel-centred and boundary placements, 200 trials per configuration and 1200 in all. The recovery criterion was frozen in advance: $T \geq 5$, peak channel within 3 of the injection, peak drift within one grid step.

The coarse-window variant recovered 0 of 500 injections at every amplitude from 4σ to 10σ under that criterion. The criterion produced that artefact and the pipeline suppresses nothing, which makes it the one withdrawn result here. The drift-matching clause cannot be satisfied on coarse windows, where every in-grid drift is sub-channel over the track, so the reported peak drift is arbitrary. Scored on detection alone the same windows recover essentially everything: an end-to-end injection through the released pipeline returns $T_\star$ of 20.5, 65.3 and 217.7 at 1, 3 and 10 times the per-integration noise, against an ideal stacking gain of $\sqrt{639} = 25$. Class B limits therefore bound carriers of any dwell fraction, with the completeness of §4.5, and what those windows lack is drift discrimination. The fine window fails the same criterion the same way, so the near-static cell there is unmeasured.

The second result is the completeness curve. On the fine window, drifting carriers ($f_{\rm drift} \geq 0.5$, pooled over grid phase and sub-channel placement) are recovered 17, 42, 58, 75 and 83 per cent at 4, 5, 6, 8 and 10σ, saturating by twice threshold power (Fig. 7). The machinery alone, with the tone injected after the baseline step, is amplitude-faithful to within −9/+14 per cent for channel-centred tones at every drift fraction. Boundary-straddling placement costs a factor 0.6–0.8 in the statistic and recovers 0 per cent at 5σ against 83 per cent for centred ones (67 and 100 per cent at 10σ), crossing 50 per cent near 1.4× threshold amplitude. Survey completeness uses the placement-pooled curve, conservative for the boundary-straddling case: ${\rm EIRP}_{50} = 1.10\,{\rm EIRP}_{5\sigma}$ and ${\rm EIRP}_{90} > 2.0\,{\rm EIRP}_{5\sigma}$, a bound only.

This is a demonstration that the pipeline recovers a carrier in a representative configuration. It is not a survey completeness function. The curve is measured on one window, and applying it to the other 118 drift-resolved windows is an assumption, bracketed as 0.5–2× in Appendix I. Table 7 says how far each of those windows sits from the calibration configuration on the axes that matter, so a reader can see which thresholds the measurement supports and which extrapolate from it. 58 of 118 windows (49 per cent) differ on at most one axis and the rest on two or more, the largest single group being the 47 that differ on two. Two axes account for almost all of the mismatch, band (55 windows) and drift-trial count (55), against 17 for channel width, 20 for integration time and 12 for beam offset. That is the useful reading for the transfer bracket: what is least well established is the transfer across band and across drift-grid size, not across integration time or position in the beam. Not spanned at all are other target environments and edge-phased drift grids. Bounding the transfer error needs the multi-configuration campaign of §6.4.

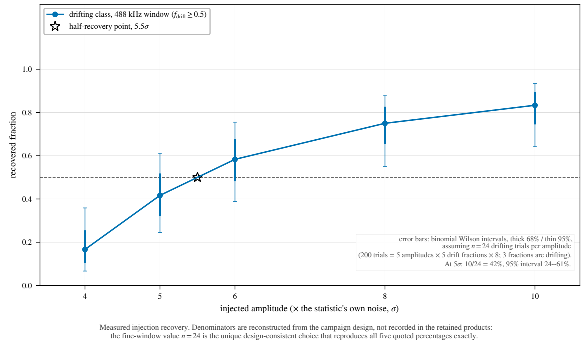

FIG. 7.— Measured injection recovery on real ALMA data, fine-channel configuration. Recovery $C(A, f_{\rm drift})$ rises with amplitude for drifting carriers; static and near-static ones fail the frozen criterion, and boundary-straddling tones can fail its peak-drift clause while still being detected. Coarse configurations are not plotted: under the same criterion they recover nothing, withdrawn in the text as an artefact of the criterion.

C. EXPLORATORY DIAGNOSTICS NOT USED FOR CANDIDATE SELECTION

Nothing in this appendix gates a disposition or derives a limit, each screen is reported whatever it returned, and the operative chain of §5.3 would read the same if none had been run.

Closure phase. The closure phase $\arg(V_{ij}) + \arg(V_{jk}) + \arg(V_{ki})$ vanishes for an unresolved point source anywhere on sky and is immune to per-station calibration errors, so it tests point-source coherence alone. A coherent far-sidelobe interferer would pass it. We scored consistency with zero on continuum detections in four locally reprocessed fields, using the resolved AT Mic Band 7 binary as a control. At these flux densities the test is useless: the median per-baseline signal-to-noise ratio is 0.5–0.7, and every field's closure phases are indistinguishable from noise, the binary included.

Chirped drift and periodicity. Stacking at constant linear drift rate is blind by design to a drift rate that changes during the observation, and to pulsed or modulated carriers. We implemented a chirped (quadratic-drift) extension and a per-channel periodicity search, both vetting the star against the eight retained control positions of that window. On the 122 retained spectra this is a demonstration and not a measurement. Both flag at the chance rate of $\sim$11 per cent, so the extension runs correctly under the null and no limit derives from it.

Cross-target frequency occupancy. With no ON/OFF cadence against radio-frequency interference, this single-pointing archival survey substitutes the many unrelated stars sharing similar correlator setups: a channel-confined feature recurring at the same observed *sky frequency* towards different targets is almost certainly interference or instrumental. The check matches crossing frequencies towards different targets when they fall within three channel widths of each other, $\sim$46 kHz. Across the 431 windows and 107 tabulated crossings no pair falls within that kernel, or within 0.5 MHz. The kernel is narrower than the β Pictoris line offsets, so those windows were dispositioned by line-catalogue inspection instead.

The null has power only where targets share frequency coverage. On the $\leq$20 pc subset 366 of 946 star pairs overlap, and 109 of those involve one of the five stars carrying formal crossings, so every crossing frequency was tested against overlapping independent targets. Shared execution blocks are excluded as identities by construction. One qualification remains. The interim product caps each window's crossing list at 50 frequencies, so 171 crossings exist against 107 tabulated, the difference being the 64 untabulated crossings of one β Pic Band 6 window. The null is demonstrated for the tabulated set and unresolved for the rest; the release ships uncapped lists.

D. PER-TARGET RESULTS

The per-window results ship as the comma-separated catalogue of the Data Availability statement. Its EIRP$_{5\sigma}$ column uses the definition of §4, on the same threshold and primary-beam correction as the continuum column, so a row's two limits are comparable; its per-channel flux-density threshold uses the carrier lane's per-channel noise rather than the tabulated `rms_mJy`, the two differing by $5 \times G(r)$. Bands 6 and 7 supply 86 per cent of the windows.

E. THE VELOCITY-SPACE LINE MASK: JUSTIFICATION AND RETROSPECTIVE APPLICATION

The one-channel line check failed the β Pictoris windows (§5.3): a native channel spans a fraction of a km s^{-1}, while circumstellar kinematics shift real emission by many channels. The replacement mask takes its tolerance from the kinematic budget of circumstellar gas: Keplerian motion where these molecules survive ($\gtrsim$10 au), $\lesssim$13 km s^{-1}; cold-gas linewidths $\lesssim$1 km s^{-1}; cold dense-gas tracers with no bulk-wind term for the dwarfs dominating the sample; and an uncorrected barycentric term bounded by $\pm$30 km s^{-1}. The result is empirically conservative: the largest attributed circumstellar component (β Pic, within 0.46 km s^{-1} of systemic) sits at 0.9 per cent of it, and *zero* crossing frequencies lie 13–50 km s^{-1} from their nearest *masked* transition (the wider catalogues are another matter: §5.3), so a $\pm$13 km s^{-1} Keplerian-only mask would have suppressed and released the same windows: the wider tolerance costs no yield and buys robustness against radial-velocity catalogue error. The list has also been checked prospectively: over the 33 islands Splatalogue returns 4,780 observed transitions of 233 carriers with $E_{\rm u} \leq 150$ K and $\log A_{ij} \geq -5$, and masking all of them would cost 74.8 per cent of the searched union against 0.95 per cent for the 17 used, excluding 19 of

TABLE 8

Selection function of the searched sample against a volume-limited reference census. Reference census: Gaia DR3 sources with $\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$ and $1/\varpi \leq 40$ pc (17566 objects; 8594, 49 per cent, carry a `teff_gspphot` estimate). Sample: the 88 stars with at least one searched window, at the distances the search itself used. Temperatures are available for 52 of the 88 stars (all 88 matched to the 168-entry ALMA-covered census, 58 by exact name; 52 of those matches carry a temperature), so the temperature panel is normalised within each column over classified objects only, with $T_{\rm eff}$ a proxy for spectral class. "Ratio" is the sample column fraction over the census column fraction: 1 means the sample tracks the reference population, > 1 over-representation.

Property	Census	Searched	Ratio
Distance (all 88 searched stars; 17566 census objects)			
$d \leq 5$ pc	48 (0.3%)	12 (13.6%)	49.9
5–10 pc	240 (1.4%)	13 (14.8%)	10.8
10–20 pc	2069 (11.8%)	18 (20.5%)	1.7
20–30 pc	5391 (30.7%)	18 (20.5%)	0.7
30–40 pc	9818 (55.9%)	27 (30.7%)	0.5
$T_{\rm eff}$ class[a] (52 searched; 8594 census)			
A ($\geq$ 7500 K)	24 (0.3%)	1 (1.9%)	6.9
F (6000–7500 K)	402 (4.7%)	9 (17.3%)	3.7
G (5300–6000 K)	860 (10.0%)	16 (30.8%)	3.1
K (3900–5300 K)	1400 (16.3%)	5 (9.6%)	0.6
M (< 3900 K)	5908 (68.7%)	21 (40.4%)	0.6
Other axes			
Known planet hosts[b]	20 (11.9%)	18 (20.5%)	1.7
Searched stars / systems	–	88 / 81	–

[a]Bins are temperature bins, not MK types: A $\geq$ 7500, F 6000–7500, G 5300–6000, K 3900–5300, M < 3900 K. Gaia `teff_gspphot` is unavailable for about half the reference census, preferentially for the coolest and faintest objects, so the census M fraction is a lower limit and the earlier-type ratios are correspondingly upper limits. White dwarfs are not separable in this parameterisation and fall wherever their (unreliable) photometric temperature places them.

[b]The Gaia reference census carries no planet information; the census column here is instead the 168-entry ALMA-covered census, of which 20 entries are known planet hosts. The 88 searched stars include 18 hosts of 41 known planets.

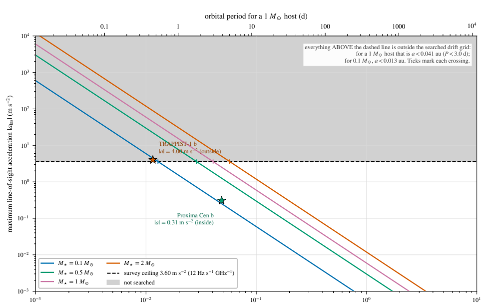

Fig. 8.— The drift-rate ceiling as a physical selection function. Maximum star–planet relative acceleration, edge-on and maximised over orbital phase, as the carrier drift rate it induces (Hz s^{-1} per GHz of carrier frequency; equivalently $\dot{\nu}/\nu = \dot{v}/c$), against orbital period, for host-star masses 0.09–1.8 $M_\odot$ (lines) and the known planets of the planet-hosting targets (points). Shaded band: the survey's drift-search ceiling, above which a transmitter lies outside the searched grid. Only TRAPPIST-1 b reaches it (§6).

TABLE 9

Per-band survey characterisation of the 431 searched windows. "Sys." counts unique stellar systems searched in the band, so a system observed in two bands is counted in both rows (81 systems, 88 stars in total); "EBs" the contributing execution blocks, attributed by the provenance of the de-duplicated windows; $\Delta\nu_{\rm ch}$ the native channel width in MHz (median, range in brackets); "Union" the unique sky frequency searched in the band after merging overlapping windows; EIRP$_{5\sigma}$ the median nominal threshold across the band's windows; $\dot{\nu}_{\rm ceil}$ the range of drift-rate ceilings in Hz s^{-1} per GHz of carrier frequency. The "All" row is computed over the whole sample, not summed down the columns: EBs and systems recur across bands and the frequency unions are disjoint by construction.

Band	Sys.	EBs	Win.	$\Delta\nu_{\rm ch}$ (MHz)	Union (GHz)	Med. EIRP$_{5\sigma}$ (W)	$\dot{\nu}_{\rm ceil}$
3	8	9	40	15.63 (0.244–15.63)	20.6	1.1×10^{14}	12.0–13.3
4	1	1	4	15.63	6.9	2.6×10^{13}	12.0
5	1	1	4	15.63	6.9	4.4×10^{13}	12.0
6	54	54	216	15.63 (0.015–31.25)	29.4	1.3×10^{15}	12.0–13.3
7	33	34	155	15.63 (0.061–15.63)	22.6	2.9×10^{15}	12.0–13.3
8	3	3	12	15.63 (0.061–15.63)	6.6	6.1×10^{16}	12.0
All	81	102	431	15.63 (0.015–31.25)	93.1	1.4×10^{15}	12.0–13.3

20 crossings, CP−72 2713 included. HCN, CS, SiO, CN and H_2CO are already masked: no coverage figure and no disposition changes.

The repaired mask. The four defects and their repair are in §5.3. The rounding defect is why the released CO(1→0) offset column, referred to the frozen mask's rounded 115.271000 GHz entry, reads 0.202 MHz (0.53 km s^{-1}) smaller than the laboratory-referred value of Table 10. A cheaper rule exists: the archive records the rest frequency each execution block was tuned to, so masking each window's own tuned line, inside an LSR-frame tube of a few $\times 10\,\mathrm{km\,s^{-1}}$, catches the β Pictoris, HD 48370 and [C I] windows automatically.

The crossing ledger, once: of the 20 windows with an on-star crossing, 4 also beat their control ring and three of those are accounted for by the mask. Two further crossing windows fall inside the mask without beating their ring (the unflagged β Pic Class A window at -0.46 $\mathrm{km\,s^{-1}}$; HD 285968 at +8.7 $\mathrm{km\,s^{-1}}$, already rejected by its ring, maximum 10.9 against a star peak of 6.5), and the remainder lie far outside any circumstellar tolerance (AU Mic +379, LHS 1140 +115 $\mathrm{km\,s^{-1}}$). The other 16 are ordinary noise crossings: in each, the window's own control ring reaches or exceeds the stellar value (ring-maximum median 5.84 against a crossing median 5.18), the expected Gaussian-tailed population of moderate 5–5.5σ crossings with comparably strong peaks among 512 controls. But the transitions that make astrophysical false positives cheap are also natural reference frequencies for a deliberate communicator (Cocconi & Morrison 1959), so the released products keep a known-line-coincident list: every on-source 5σ crossing inside the masked intervals, with its offset from each nearby transition.

F. THE ANCILLARY CONTINUUM LANE

The continuum lane is an exploratory by-product: 57 measurements, 17 detections and 40 limits. A millimetre excess is at best an indirect and model-dependent observable of waste-heat technology. An "upper limit" here is 5× the primary-beam-corrected map rms at the star's position, and the deepest non-detection is towards TRAPPIST-1 in Band 3. Under the combined error model sixteen of the seventeen detections stay above 5σ, the exception being η Corvi Band 7 at 4.7σ.

Every detection is astrophysically unsurprising: the unresolved UV Ceti binary; six systems of disc or chromospheric emission (β Pic, AU Mic, AT Mic and τ Cet among them); six stars at photospheric flux; and two late-M dwarfs of documented magnetic activity, LP 349−25 and the auroral emitter LSR J1835+3259 (Hallinan et al. 2015). Closure phase discriminates structure for none of them at these flux densities.

The anomaly screen. Each star's flux is tested by a robust median/MAD statistic against its own photospheric expectation and against same-$T_{\rm eff}$ stars in coarse bins of at least five. The error model, fixed before any flag, adds the map noise, a calibration term of 5 per cent in Bands 3–6 and 10 per cent in Bands 7–8, and a model term propagated from the Gaia temperature and radius errors. It has *no intrinsic-variability term.* That is adequate for a photosphere and wrong for an active M dwarf, whose millimetre flux can vary by factors of several, so the screen is uninformative for exactly the stars a radio-stellar reader cares about. It flags for inspection and tests no per-star hypothesis. One outlier results, χ^1 Ori at Band 3, consistent with a spectroscopic G0V binary's unresolved companion or a chromosphere.

G. FALSE-ALARM ACCOUNTING UNDER THE SYMMETRIC CRITERION

The per-window rate is 1/513 and the survey expectation 431/513 = 0.84 flags. The floor falls only if the control ensemble grows, which the primary beam bounds. Windows are not independent trials either, grouping into 102 execution blocks that share weather, calibration and correlator setup, and that clustering is measured here rather than assumed. Under exchangeability the designated position may be moved to any control, so we draw one index per execution block and apply it to all of that block's windows. A window is then rank-first exactly when the drawn index is the index of its own control maximum, so the distribution follows in closed form: mean 0.84 rank-first windows, $P(\geq 4) = 0.011$ and $P(\geq 5) = 1.7 \times 10^{-3}$, against the independent-Poisson 1.7×10^{-3}. Clustering changes so little because the *position* of the control maximum is not shared: 1 of the 805 within-block window pairs put their maxima at the same control, against 1.6 by chance. Under the 5σ gate the same calculation expects 0.24 stage-1 outliers and gives the single unattributed one $P(\geq 1) = 0.22$. The survey maximum over 20,000 clustered draws has median 9.9 and 5th to 95th percentiles 5.7–19.8, so a maximum the size of CP−72 2713's arises in most draws.

The trials budget, in one set of units. A window holds 177–6 866 208 channel×drift cells (median 468), and the survey 4.60×10^7 in total. A one-sided Gaussian 5σ tail (2.87×10^{-7}) therefore predicts 13.2 chance cells across the survey, an expectation in cells and not in windows. The frozen products carry the observed count in the same units: 229 on-star cells reach 5σ, in 20 windows, and 174 of them belong to the four stage-1 outliers, three of which are CO. The remaining 55 cells, over 16 windows, stand against the 13.2 expected. That factor of 4.2 is a product of two, and only the second is correlation. Correlation cannot move the expected *number* above 5σ, which is Np whatever the correlation; it moves the clumping. The two parts are the window count and the cell multiplicity inside a crossing window, the second being channel adjacency. The budget is also almost entirely fine-class, which is why all 20 crossing windows are fine ones.

The control ensembles calibrate this empirically, no Gaussian assumption surviving. The same arithmetic on each window's 512 controls predicts 134 of the 431 windows carrying at least one control above 5σ; 125 do, agreeing to 7 per cent. The observed per-window control maxima track the iid-Gaussian prediction for $512 \times n_{\rm cells}$ trials at a median ratio of 0.995 (5th–95th percentile 0.91–1.12), separately in both classes: fine median observed control maxima 5.77 against an expected maximum 5.74, coarse 4.41 against 4.47.

That agreement is partly bookkeeping, because $n_{\rm cells}$ counts grid cells and not independent trials, so the arithmetic follows rather than an asserted correction. On the frequency axis ALMA's online Hanning smoothing correlates neighbouring channels at $\rho_1 = 2/3$ and $\rho_2 = 1/6$ (Appendix A), so the variance of a sum over n channels exceeds n by $1 + 2\rho_1 + 2\rho_2 = 2.67$ and that many grid channels carry one channel's worth of independent information. On the drift axis the grid is deliberately oversampled at two steps per channel of traverse over the track, so adjacent trials share nearly all their integrations and count 2 for one. The axes are independent of each other, so the combined over-count is $2.67 \times 2 = 5.3$. Dividing $n_{\rm cells}$ by it lowers the predicted maxima from 5.74 to 5.45 in the fine class and from 4.47 to 4.10 in the coarse one, against 5.77 and 4.41 observed: a residual +6 and +8 per cent, in the two classes independently. Mild non-Gaussianity of that size is exactly why the operative statement is the empirical rank and not the Gaussian budget.

The same correction disposes of most of the on-star cell excess. Of the factor 4.2 above, the part contributed within a crossing window is 3.4 cells observed against 1.2 expected under independence, a factor 2.8, which is what an over-count of 2.67 on the frequency axis predicts: one real threshold excursion is counted about three times. What that leaves is the window-level factor, 16 windows carrying at least one unattributed on-star crossing against 10.9 expected, a factor 1.5 whose one-sided Poisson probability is 0.09. Once the four dispositioned windows are set aside, the residual excess is consistent with chance at that level. It is not zero, runs one way, and carries the same sign at the controls, so we report it as an open systematic. The crossing list is dominated by CO(2→1) at 230.5 GHz toward disc hosts, and that a residual excess of on-star chance crossings at the tens-of-per-cent level cannot be excluded. The operative false-alarm statement remains the rank against the control ensemble.

The factor of 16 fixes what a single window can ever mean here. The rank floor is $1/(N_{\rm ctrl}+1) = 1/513 = 1.95\times10^{-3}$; the survey-wide Bonferroni scale is $0.05/431 = 1.2 \times 10^{-4}$; their ratio is 16. Correlation widens it. With $N_{\rm eff} \simeq 30$–43 independent spatial trials the ensemble can really resolve only 0.023–0.032, and the gap to the Bonferroni scale becomes a factor 196 to 278. Drawing more controls would not help, the primary beam bounding the annulus. This is the design's central limitation, and it is why authentication is left to recurrence and astrophysical attribution. That figure is the compact case: taking the archive's own synthesised beam (0.10–5.7 arcsec, median 0.84) and $N_{\rm geom} = \pi(r_{\rm out}^2 - r_{\rm in}^2)/1.133\theta_{\rm beam}^2$ gives 36 to 1.4×10^5 resolution elements, median 995, across the 431 windows. That counts what the annulus *contains*, so it caps $N_{\rm eff}$ at the 512 positions actually drawn and shows the blanket 30–43 to be a worst case, not a typical one. The same metadata exposes two geometric limitations, unrepaired here because repairing them means re-running the extraction. 141 windows (33 per cent) come from 38 ACA 7 m execution blocks while $\theta_{\rm PB}$ is $1.22\,\lambda/12\,$m throughout, so there the annulus sits at a smaller fraction of the true primary beam than 0.14–0.78 implies and its gains are higher than quoted; and in 133 windows $r_{\rm in}$ falls below the synthesised beam, so the innermost controls are not resolution-independent of the star, which for a typical ACA geometry is ∼4 per cent of the 512, far below the median's breakdown point. Of the four stage-1 outliers only HD 48370 Band 6 is ACA. Both consequences run in the conservative direction. On the true 7 m beam the ACA annuli span 0.08–0.46 of $\theta_{\rm PB}$ at a median area-weighted gain of 0.75 against 0.47 for the 12 m windows, so exchangeability in flux density improves there; and of the 41 windows carrying a primary-beam correction above 2 per cent, 36 are 12 m and 5 are 7 m, with all four at the 1.81× maximum 12 m, so the applied correction is overstated only for the five ACA rows, where the 1.71× dish-diameter ratio makes a ×1.05 correction really ×1.017.

Dropping the ACA windows tests what that geometry costs. The 290 12 m windows alone cover 62 stars in 55 systems over 64 blocks, carry 8 crossings and 3 rank-first windows against 0.57 expected, hold 3 of the 4 stage-1 outliers, the one lost being HD 48370's foreground CO, and keep uniform stellar ranks ($D = 0.050$, $p = 0.44$). The deepest trigger and the deepest $P_{\rm eff}$ are 12 m windows and do not move, so no survey-level statement changes. The ACA stratum alone departs mildly ($D = 0.13$, $p = 0.018$), opposite in direction to what a mis-scaled annulus gain would produce.

Excluding the 15 β Pictoris windows leaves one unattributed crossing among 416, against 0.81 expected ($P(\geq 1) = 56$ per cent).

H. ANALYSIS PROVENANCE AND ROBUSTNESS CHECKS

This appendix holds the analysis history: how the search statistic came to be what it is, the checks run on the windows the old statistic flagged, the frame audit behind the CO attributions, the line-mask repair, and the entries excluded from the released catalogue. None of it changes a number in the main text.

CHRONOLOGY OF THE STATISTIC REVISION

An earlier region-max statistic maximised over an $n_{\rm src} = 135$-position region centred on the star against 512 single-position controls, and under it a crossing towards AU Mic in Band 6 at 230.83 GHz was carried as a candidate. The released repository dates both steps. That flag is first committed on 2026 August 31 (`8e00f90e8f8c`), in the manuscript of the day and in its machine-readable aggregate, and the symmetric form of Eq. 2 is adopted as operative 9 days later, on 2026 September 9 (`0c465f2661bd`). The redesign therefore postdates the flag. Its justification is algebra that owes nothing to these data, since under exchangeability an $n_{\rm src}$-position maximum at the star against single-position controls ranks first with probability $n_{\rm src}/(n_{\rm src} + 512) = 20.9$ per cent, which the symmetric form removes.

Five pre-freeze changes are logged with their effect on released products in the repository change-log. Only this one changes a released number. Counted *without* the $\geq 5\sigma$ amplitude gate, that is on rank alone, rank-first windows 65/431 (15.1 per cent) become 5/431 (1.2 per cent). Counted *with* the gate, which is the stage-1 spatial outlier of the main text, 7 become 4 and AU Mic ceases to be one. Nothing derived from the retired statistic enters an operative false-alarm probability.

AU MIC: THE THREE TESTS IN FULL

All three converge on the same conclusion: the AU Mic crossing is not a candidate under the symmetric statistic, it would not have been one on the original data either, and nothing about the reclassification required data that did not already exist when the window was first flagged.

The within-window add-one rank of the AU Mic crossing is $p = 0.021$, a different quantity from the survey-wide rate at which any window's ring maximum reaches 5.97, $p = 0.081$, and both are far from the 1/513 floor. Under the three treatments that decide the crossing, the frozen region-max products, a fully symmetric reprocessing of the same first-epoch block, and the second-epoch block of the same programme, the star statistic is 5.97, 4.93 and 4.77 against control maxima of 5.85, 7.70 and 6.78, with add-one star ranks $p = 0.81$ and 0.97 in the latter two.

Localisation. Reprocessed from the raw science data model with the observatory pipeline's own calibration recipe (310 integrations against the release pipeline's 249, on a finer drift grid) and re-run through the symmetric statistic of §4.2, the crossing window's star does not beat its control ensemble. The frozen products agree only asymmetrically: the source-region peak exceeds the 512-control maximum by 0.12, below that statistic's window-to-window scatter, and the crossing fails the single-position variant too.

Recurrence. The one further public Band 6 block covering 230.83 GHz, the same programme executed 2014 June 5, was searched identically. No channel within ±10 of the first-epoch flagged channel exceeds 3.1, and nothing is localised to the star. Its band rms (1.48 mJy per 488-kHz channel) matches the first epoch's 1.9 mJy, so a comparable excess would have been recovered, and the frequency recurs at no other target. Intermittent or precisely-scheduled transmissions outside the sampled windows stay unconstrained.

Statistical scale. The on-source peak (5.97) sits at the 92.1th percentile of the control-maxima distribution over all 431 windows, 34 of which (7.9 per cent) match or exceed it, add-one empirical $p = 0.081$ before any multiple-testing correction. The crossing matches no catalogued molecular transition (Table 10), and closure-phase vetting finds it point-source consistent, which is one-sided evidence (Appendix C). Two astrophysical contexts are recorded for re-examination and neither grounds the rejection. AU Mic is an exceptionally active flare star, but the block's own quiescent continuum (0.15 mJy) rules out the later-epoch GHz-wide flares of MacGregor et al. (2020) (16.8 and 6.1 mJy), since an 11 mJy channel-confined excess would sit $\sim$70× above a continuum that shows nothing. And the controls share the primary-beam footprint, so smooth structure such as the debris belt raises star and controls alike and cannot by itself produce a localised excess.

THE R_σ RESIDUAL-SCALE TEST

For each of the four stage-1 windows the pipeline's own baseline-subtracted per-integration residuals at the star and at the 8 retained control positions are standardised by its noise map $\sigma(t,\nu)$ and reduced to a robust scale over the (integration, channel) plane, the crossing channels excluded at every position alike: $R_\sigma = \sigma_{\star,\mathrm{resid}}/\mathrm{median}(\sigma_{\mathrm{ctrl,resid}})$ is 1.000 (CP−72 2713), 0.995 (HD 48370), 1.001 (β Pic B3) and 1.001 (β Pic B6), the one star measurably *quieter* than its ring being HD 48370, whose ring is full of CO. Eight draws set the position-to-position scatter, hence a bound at the per cent level and not a tenth of one, and their own spread is 0.6 per cent; retaining all 512 control spectra (§6.4) would fix that and the test's blindness to a channel-confined artefact together.

THE FULLY SYMMETRIC REGION-MAX CHECK

A fully symmetric region-max variant applies identical source-region, drift-rate and frequency maximisation at the star and at every control. It needs products the frozen release does not carry, so its domain is seven locally reprocessed windows. All seven give $T_\star = 3.7$–4.9 against control maxima of 6.7–7.7, $p = 0.81$–1.00, so the star is exchangeable with its ring wherever the test runs at full symmetry. The single-position calibration of §5.3 remains the operative release-wide criterion and the two are never interchanged (Appendix G).

THE β PICTORIS FRAME AUDIT

Table 10 carries the frame chain for the line-attributed windows. It recovers the whole check list untuned: frequencies to the frame chain's stated tolerance, velocity concordance across epochs and transitions, localisation against the control ensemble, and drift class zero to within the grid.

Two further properties fit CO and not a carrier. The stage-1 Band 6 window carries 114 formal $\geq 5\sigma$ crossings where a channel-confined carrier admits one. The retained products do not record whether those channels are contiguous, so the count sets a floor of 2.26 km s^{-1} on the feature's extent without measuring it. The peak flux densities, 54 and 78 mJy per channel, are the right order for this belt's spatially integrated CO. They are per-beam peaks on emission resolved in both beams, and Matrà et al. (2017) find the gas subthermally excited, so beam dilution and excitation both depress their ratio below optically thin LTE.

Stellar emission is a competing explanation for any stage-1 outlier towards an active young star, and the search suppresses it twice. Gyrosynchrotron and free-free emission from chromospheres and coronae are broadband, while the block-median baseline of §4.2 removes everything broader than its 65-channel passband, $\sim$32 MHz at the fine class, a fractional bandwidth of $\sim 10^{-4}$; MacGregor et al. (2020)'s AU Mic flares are coherent across the full 8 GHz. Flares are also time-confined, so the inverse-variance stack over all integrations dilutes them. What survives is the multiplicative route of §4.2, a flare continuum times a spectral calibration residual, bounded there and measured for CP−72 2713 in §5.3. The search is blind to the time axis by construction, and the retained dynamic spectra are the only handle on it.

HD 48370 in Band 8. The carbon tuning says the same thing by omission. All twelve Band 8 windows here belong to η Crv, HD 48370 and HD 61005 and are [C I] $^3P_1$–$^3P_0$ (492.160651 GHz) tunings. HD 48370's carries a 5.05σ on-star crossing against a ring maximum of 6.23, so it is not one of the four stage-1 outliers. Its crossing channel sits -129 km s^{-1} from [C I], so it is not [C I] emission from the disc, the cloud or the star, and its own ring dispositions it

TABLE 10
LINE-ATTRIBUTION AUDIT THROUGH THE FRAME CHAIN OF §4.3: THE THREE LINE-ATTRIBUTED WINDOWS, AND THE CLASS A β PICTORIS BAND 6 WINDOW THAT IS NO STAGE-1 OUTLIER YET SHOWS THE SAME CO COMPONENT. TOPOCENTRIC OFFSETS ARE OBSERVED MINUS CATALOGUED LABORATORY REST FREQUENCY (PICKETT ET AL. 1998), SO A NEGATIVE OFFSET MEANS THE OBSERVED FREQUENCY LIES BELOW REST. THE BARYCENTRIC CORRECTION IS THE EARTH-EPHEMERIS TERM AT THE EXECUTION-BLOCK MID-TIME. THE AU MIC SYSTEMIC VELOCITY IS CATALOGUED (FOUQUÉ ET AL. 2018). FOR β PICTORIS WE ADOPT $v_{\rm sys} = +20.0 \pm 0.7\,{\rm km\,s^{-1}}$ (MATRÀ ET AL. 2017) RATHER THAN THE *Gaia* DR3 VALUE OF $+16.84\,{\rm km\,s^{-1}}$ (GAIA COLLABORATION ET AL. 2022), BECAUSE THE STAR IS AN A6V ROTATOR INSIDE THE HOT-STAR TEMPLATE REGIME IN WHICH DR3 RADIAL VELOCITIES CARRY A KNOWN SYSTEMATIC (BLOMME ET AL. 2023).

Flag	UTC start	$f_{\rm obs}$ (GHz)	Transition	Topo. offset (MHz; $\rm km\,s^{-1}$)	Bary. corr. ($\rm km\,s^{-1}$)	$v_{\rm sys}$ ($\rm km\,s^{-1}$)	Stellar-frame offset ($\rm km\,s^{-1}$)
β Pic B3	2022-03-03 21:49	115.260644	CO(1→0)	-10.56; −27.46	−7.90	+20.0	+0.44
β Pic B6	2019-03-12 00:14	230.516128	CO(2→1)	-21.87; −28.44	−8.15	+20.0	-0.29
AU Mic B6	2014-08-18 04:01	230.825230	CO(2→1)	+287.23; +373.51	−10.02	−4.71	+378.82
β Pic B6†	2013-10-06 06:23	230.528134	CO(2→1)	-9.87; -12.83	+7.63	+20.0	-0.46

†Class A Band 6 window of a different execution block, without a stage-1 flag (Appendix D), shown because it independently shows the same CO component.

as an ordinary noise crossing. What the window does show is no line at the star in this disc's carbon tuning, which is what Cataldi et al. (2023) report. The crossing peak in the flagged window itself is 0.82 Jy in one 122-kHz channel with the ring at the same level.

THE LINE MASK AS EXECUTED, AND ITS REPAIR

Four defects in the mask as executed are all repaired in §5.3. Its rest frequencies were rounded to 1 MHz, worth up to $1.59\,{\rm km\,s^{-1}}$. It carried no atomic carbon, though every Band 8 window here is a [C I] tuning, and no hydrogen recombination line, though H30α falls in the standard Band 6 tuning of much of the sample. It was also applied in the stellar frame alone, the wrong frame for the Galactic foreground behind one of the three line attributions. The repaired catalogue adds [C I] at 492.160651 GHz and H30α at 231.900928 GHz, against 15 transitions of 9 species before. The species count also reconciles the text with the products, which always used 9 species; earlier versions of this paper said eight, omitting H_2CO.

Re-application uses each crossing channel's own frequency as the pipeline recorded it, and no disposition changes. The two β Pictoris crossings remain inside the stellar-frame tube. HD 48370's remains inside it at $-17.8\,{\rm km\,s^{-1}}$ and is now caught by the LSR-frame tube as well, at $-23.9\,{\rm km\,s^{-1}}$, the frame its foreground attribution always implied. CP−72 2713's lies $-1323.2\,{\rm km\,s^{-1}}$ from the nearest masked transition in the topocentric frame and $-1300\,{\rm km\,s^{-1}}$ in the LSR frame, so no tube of this mask excludes it; a wider catalogue does place entries inside the tube, which §5.3 treats on the physics. Across all 20 crossing windows the repaired catalogue moves the nearest transition in exactly 1 case, HD 48370's Band 8 crossing at 491.949074 GHz, whose nearest entry changes from CO(4→3) $20098\,{\rm km\,s^{-1}}$ away to [C I] at $-129\,{\rm km\,s^{-1}}$. That is still outside the tube, and the frame corrections are at most a few tens of $\rm km\,s^{-1}$, so no frame masks it; it was never an outlier, its own control ensemble exceeding it. The two new transitions cost a further 0.24 GHz of unique searched frequency, 0.3 per cent.

ENTRIES EXCLUDED FROM THE RELEASED CATALOGUE

The four classes of excluded entry. (i) Four target/bands (Barnard's Star [B7], Wolf 359 [B7], SCR J1845−6357, Wolf 358) have no carrier result from a process-level failure; each keeps a valid continuum non-detection and no EIRP threshold. (ii) Giclas 9−38A and Giclas 9−38B [B3] are excluded entirely, the pair's only matched MOUS pointing ∼52 arcmin from either star, 35× its own primary-beam FWHM; the carrier step aborts correctly, continuum imaging lacks that guard. (iii) Six windows across five stars (51 Eri, HD 10647, HD 139664, TWA 7, HD 92945 twice) fail the noise test of §5.2: all are 15.62-MHz coarse windows whose reported rms is inconsistent with their reported integration by three to four orders of magnitude and sits a factor 388–669 below the shallowest sibling of the same block and channel width, a normalisation signature, not weather or configuration (§5.2). (iv) The catalogue entry "γ Lupi" resolves to the F4V debris-disc host HD 139664 at 17.40 pc, not the naked-eye γ Lupi; all such entries are relabelled and no measured value changes. The supporting quality audit is clean across all 431 windows bar four χ^1 Ori Band 3 windows flagged as elevated (0.9–1.1× the Band 3 median, weakening their thresholds by $\lesssim$20 per cent in amplitude).

I. ILLUSTRATIVE CONDITIONAL DETECTION-PROBABILITY CALCULATION

No number here appears among the headline quantities of the paper, and the appendix is kept for the framework alone. **The fraction below is conditional on the searched frequencies, epochs and morphology, and quoting it outside that conditioning would misstate it by orders of magnitude.** A zero detection constrains a fraction of systems only through a per-system probability, factorised as

$$C_i(P) \;=\; F_i\; R_i(P)\; D_i\; C_{{\rm morph},i}, \qquad \text{(I1)}$$

with F_i the chance the transmitter's sky frequency lies in system i's searched intervals, $R_i(P)$ the chance a transmitter of EIRP P would have crossed that system's threshold, D_i the chance it emitted during the analysed archival interval, and $C_{{\rm morph},i}$ the measured recovery of a carrier of the searched drift and temporal morphology (Appendix B). The

TABLE 11
ILLUSTRATIVE CONDITIONAL DETECTION-PROBABILITY CALCULATION. *These numbers assume every searched star was broadcasting continuously, in band, at the stated power during the observation, and they are not a demographic limit on any stellar population.* (A) THE 95 PER CENT UPPER LIMIT $f_{95}(P)$ ABOVE EIRP P, WITH n THE SYSTEMS OF NON-ZERO COMPLETENESS THERE. (B) EPOCH-ACTIVITY DEPENDENCE AT $P \geq 10^{16}$ W.

(a)

P (W)	thresholded (unit C)		measured C	
	n	f_{95}	n	f_{95}
3×10^{13}	4	52.7%	2	unconstrained
10^{14}	16	17.1%	11	47.1%
10^{15}	59	5.0%	47	9.5%
10^{16}	80	3.7%	58	6.3%
10^{17}	81	3.6%	59	6.0%

(b)

Epoch activity p_{epoch}	f_{95} thresholded	f_{95} measured
1 (continuous)	3.7%	6.3%
0.5	6.5%	11.0%
0.1	29.2%	48.8%
0.01	unconstrained	unconstrained

probability of zero credible detections if a fraction f of systems hosts one is

$$P(N_{\mathrm{det}} = 0 \,|\, f) = \prod_{i=1}^{N_{\mathrm{sys}}} (1 - f\, C_i)\,, \tag{I2}$$

and the one-sided 95 per cent upper limit f_{95} solves $P(N_{\mathrm{det}} = 0 \,|\, f_{95}) = 0.05$ over the 81 systems.

Everything below sets $F_i = D_i = 1$ *by conditioning, not by measurement.* At EIRP $\geq 10^{16}$ W the measured form gives $f_{95} \approx 6$ per cent over the 58 systems carrying a Class A window. Folding in the dwell-measured Class B curve extends the calculation to 80 systems and moves the number to 4.3 per cent. Table 11 carries the ladder in EIRP and in epoch activity. How far this travels is bounded three ways. The Class A recovery curve is measured on one configuration and applied to all 118 Class A windows (Table 7); scaling it by 0.5–2× moves the number to 5.0–12.3 per cent, which is the honest width. The C_i are not independent, windows grouping into 102 execution blocks with an intraclass correlation of 0.24, but that design effect is an order below the transfer term. And under any transmitter-frequency prior spread across ALMA's tuning range the bound vanishes, the per-system Class A union being 1.73 GHz in the median against ∼75 GHz.

Writing p_{epoch} for the chance a transmitter is on in any one epoch, a system *searched* in n_i independent execution blocks is sampled n_i times, so $D_i = 1 - (1 - p_{\mathrm{epoch}})^{n_i}$. 18 of the 81 systems have $n_i > 1$, against 63 with two or more public blocks archived, so the table reports what the searched epochs support. With at most three epochs at year-scale separations this is a phase-averaged bound, and an intermittent transmitter is at least as plausible *a priori*, so the $p_{\mathrm{epoch}} = 1$ entries should never be quoted without it.

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